DAMIAO | 达妙科技

Shenzhen Damiao Technology Co., Ltd.
2026 Product Selection Manual

Professional Joint Modules, Intelligent Motion Control

Contents

Contents 3
Motor Module Selection Guide 5
Product Capabilities 7
Motor Model Naming Convention 8
Joint Motor
DM-J3507-2EC 9
DM-J3507-2EC (48V) 10
DM-J4310-2EC V1.1 11
DM-J4310-2EC V1.1（48V） 12
DM-J4310-2EC V1.2 13
DM-J4310-2EC V1.2（48V） 14
DM-J4310P-2EC 15
DM-J4310P-2EC （48V） 16
DM-J4340-2EC 17
DM-J4340-2EC（48V） 18
DM-J4340-2EC V1.1 19
DM-J4340-2EC V1.1（48V） 20
DM-J4340P-2EC 21
DM-J4340P-2EC（48V） 22
DM-J4340P-2EC V1.1 23
DM-J4340P-2EC V1.1（48V） 24
DM-J6006-2EC 25
DM-J6248P-2EC 26
DM-J8006-2EC V1.1 27
DM-J8009-2EC 28
DM-J8009P-2EC 29
DM-J8009P-2EC V1.1 30
DM-J10010-2EC 31
DM-J10010L-2EC 32
DM-J10422P-2EC 33
DM-J8520P-2EC 34
DM-J6216L-2EC 35
DM-J6216P-2EC 36
DM-J4310M-2EC 37

Contents

Separable Motor

DM-S3519-1EC 38

DM-S2325-1EC 39

Hollow Gimbal

DM-G6220 40

DM-G6220 （48V） 41

DM-G8518 42

Hub Motor

DM-H6215 43

DM-H3510 44

DM-H65 45

DM-H55 47

Harmonic Drive Joint Motor

DM-JH11-51-2EC 49

DM-JH11-101-2EC 50

DM-JH14-51-2EC 51

DM-JH14-101-2EC 52

DM-JH17-51-2EC 53

DM-JH17-101-2EC 54

DM-JH20-51-2EC 55

DM-JH20-101-2EC 56

Direct Drive Motor

DM-D5730-1EC 57

DM-D5730-1EE 58

Driver

Driver 59

Development Kit

Motor Setup Kit & Configuration Software 60

Development Board 61

NX Carrier Board 62

IMU Module 63

Motor Module Selection Guide

Type	Motor Model	Rated Torque (Nm)	Peak Torque (Nm)	No-load Speed (rpm)	Rated Speed (rpm)	Gear Ratio	Dimensions: Diameter *Height(mm)	Weight (g)	Rated Power (W)	Drive Configuration
Joint Motor	DM-J3507-2EC	0.8	3	460	150	7	46*37.9	≈ 150	12	Integrated
	DM-J3507-2EC(48V)	0.8	3	900	150	7	46*37.9	≈ 150	12	Integrated
	DM-J4310-2EC V1.1	3	7	200	120	10	57*46	≈ 306	37	Integrated
	DM-J4310-2EC V1.1 (48V)	3	7	400	120	10	57*46	≈ 306	37	Integrated
	DM-J4310-2EC V1.2	3.5	12.5	200	120	10	57*46	≈ 306	44	Integrated
	DM-J4310-2EC V1.2 (48V)	3.5	12.5	450	120	10	57*46	≈ 306	44	Integrated
	DM-J4310P-2EC	3.5	12.5	200	120	10	57*49.05	≈ 330	44	Integrated
	DM-J4310P-2EC (48V)	3.5	12.5	450	120	10	57*49.05	≈ 330	44	Integrated
	DM-J4340-2EC	9	27	52.5	36	40	57*53.3	≈ 362	33	Integrated
	DM-J4340-2EC (48V)	9	27	100	36	40	57*53.3	≈ 362	33	Integrated
	DM-J4340-2EC V1.1	12	40	56	36	40	57*53.3	≈ 362	45	Integrated
	DM-J4340-2EC V1.1 (48V)	12	40	112	36	40	57*53.3	≈ 362	45	Integrated
	DM-J4340P-2EC	9	27	52.5	36	40	57*56.5	≈ 381	33	Integrated
	DM-J4340P-2EC (48V)	9	27	100	36	40	57*56.5	≈ 381	33	Integrated
	DM-J4340P-2EC V1.1	12	40	56	36	40	57*56.5	≈ 381	45	Integrated
	DM-J4340P-2EC V1.1 (48V)	12	40	112	36	40	57*56.5	≈ 381	45	Integrated
	DM-J6006-2EC	4	11	226@24V 408@48V	150	6	76*36.5	≈ 335	62	Integrated
	DM-J6248P-2EC	30	97	60@24V 115@48V	40	48	76*62.5	≈ 660	125	Integrated
	DM-J8006-2EC V1.1	8	20	194.2@24V 392.8@48V	120	6	96*40	≈ 559	100	Integrated
	DM-J8009-2EC	20	40	168@24V 335@48V	100@24V 200@48V	9	98*61.7	≈ 896	209@24V 418@48V	Integrated
	DM-J8009P-2EC	20	40	168@24V 335@48V	100@24V 200@48V	9	98*61.7	≈ 963	209@24V 418@48V	Integrated
	DM-J8009P-2EC V1.1	20	40	168@24V 335@24V	100@24V 200@48V	9	98*61.7	≈ 950	209@24V 418@48V	Integrated
	DM-J10010-2EC	40	150	75@24V 150@48V	50@24V 100@48V	10	112*62 (Max. overall demension: 120)	≈ 1485	209@24V 418@48V	Integrated
	DM-J10010L-2EC	40	120	100@24V 200@48V	70@24V 100@48V	10	120*53 Height (Excluding Dowel Pin)	≈ 1375	293@24V 418@48V	Integrated
	DM-J10422P-2EC	100	400	120	100	22	110*89.9	≈ 2700	1047.2	Integrated
	DM-J8520P-2EC	50	176	124	120	20	90*70	≈ 1392	628	Integrated

Motor Module Selection Guide

Type	Motor Model	Rated Torque (Nm)	Peak Torque (Nm)	No-load Speed (rpm)	Rated Speed (rpm)	Gear Ratio	Dimensions: Diameter *Height(mm)	Weight (g)	Rated Power (W)	Drive Configuration
Joint Motor	DM-J6216L-2EC	10	45	260	200	16	67*62.5	≈ 740	209	Integrated
	DM-J6216P-2EC	10	45	260	200	16	67*62.5	≈ 740	209	Integrated
	DM-J4310M-2EC	5.2	18	232	120	10	57*53.3	≈ 370	65	Integrated
Separable Motor	DM-S3519-1EC	3.5	7.8	435	395	19.2032	42*91.5	≈ 396	144	separate
	DM-S2325-1EC	1.35	5	560	380	25	28*77.6	≈ 172.2	53	separate
Hollow Gimbal	DM-G6220	1.3	2.7	300	110	1	68*41.5	≈ 493.5	14	Integrated
	DM-G6220（48V）	1.3	4.9	650	110	1	68*41.5	≈ 493.5	15	Integrated
	DM-G8518	1.9	8.78	1290@24V 2590@48V	1000@24V 2000@48V	1	96*52.5	≈ 831	199@24V 397@48V	Integrated
Hub Motor	DM-H6215	1	2	320	120	1	68*42.5	≈ 360	12	Integrated
	DM-H3510	0.1	0.454	1700	500	1	42*31.5	≈ 116	5	Integrated
	DM-H65	6	21.5	260@24V 395@36V 480@48V	120	1	171*66.3	≈ 3150	250（6Nm 400rpm@48V S2 Duty）	external
	DM-H55	3.5	9	587@24V 883@36V 1185@48V	350	1	141*66.9	≈ 2100	128	external
Harmonic Drive Joint Motor	DM-JH11-51-2EC	3.2	7.8	60	40	51	52*60	≈ 345	13	Integrated
	DM-JH11-101-2EC	4.8	10.5	30	20	101	52*60	≈ 345	10	Integrated
	DM-JH14-51-2EC	5.4	18	60	40	51	70*70	≈ 775	22	Integrated
	DM-JH14-101-2EC	7.8	28	30	20	101	70*70	≈ 775	16	Integrated
	DM-JH17-51-2EC	16	24	60	40	51	80*70	≈ 985	67	Integrated
	DM-JH17-101-2EC	24	54	30	20	101	80*70	≈ 985	50	Integrated
	DM-JH20-51-2EC	32	69	60	40	51	90*73	≈ 1428	134	Integrated
	DM-JH20-101-2EC	32	102	30	20	101	90*73	≈ 1434	67	Integrated
Direct Drive Motor	DM-D5730-1EC	0.65	1.9	1500	1100	1	57*95.2	≈ 628	74	Integrated
	DM-D5730-1EE	0.65	1.9	1500	1100	1	57*95.2	≈ 628	74	Integrated

Product Capabilities

Product Capabilities

Pole Pair Identification

Automatically detects motor pole pairs for quick commissioning.

Automatic Phase Detection

Automatically detects phase wiring to prevent motor damage from incorrect connections.

Parameter Identification

Automatically measures resistance,inductance, flux linkage, and rotor inertia, with self-tuning of control parameters.

Dual Temperature Protection

Independent temperature monitoring for both motor and drive.

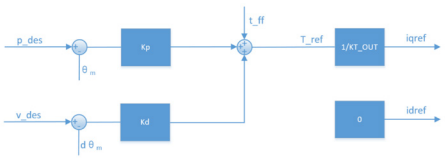
Adaptive Load Control

Dynamically adjusts control parameters to optimize performance under varying loads

Control Mode

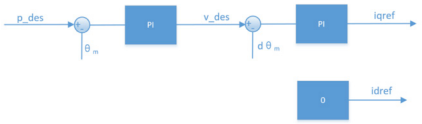
① MIT Mode

Compatible with the standard MIT control method.
Supports smooth switching and allows setting limits for position, speed, and torque (P_MAX, V_MAX, T_MAX).
CAN commands are converted into torque values.
The torque is then used as the reference for current control.



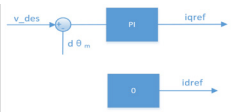
② Position Velocity Mode

Uses three control loops: position, velocity, and current(torque).
The position loop sets the target for the speed loop. The velocity loop then sets the target for the current loop.
See block diagram right.



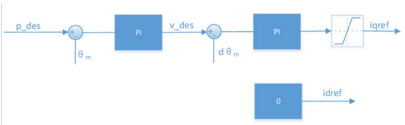
③ Velocity Mode

Keeps the motor running at the target speed.
See block diagram right.



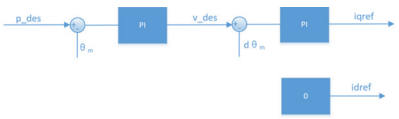
④ Force-Position Hybrid Mode

Based on position and speed control, with additional torque adjustment.
Allows control of both position and output torque at the same time.
See block diagram right.



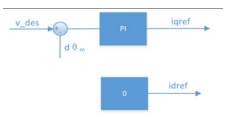
⑤ Cyclic Synchronous Position Mode (CSP)

In CSP mode, absolute position commands are transmitted periodically (the "cycle") via the fieldbus.
The controller does not generate motion profiles and directly follows the commanded position values.



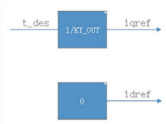
⑥ Cyclic Synchronous Velocity Mode (CSV)

In CSV mode, the host controller periodically transmits target velocity commands to the drive.
Velocity and torque control are handled internally by the driver.
See block diagram right.



⑦ Cyclic Synchronous Torque Mode (CST)

In CST mode, torque commands are sent periodically from the host controller. Torque control is handled internally by the driver.
When speed reaches the limit, the system enters speed regulation.
See block diagram right.



Motor Model Naming Convention

01 Common Naming Convention (J/S/H/G/D Series)

DM - J 43 40 X - 2E C VX.X （48V)

(Brand) DAMIAO

Product Series

J = Joint Motor
S = Separate Drive Motor
H = Hub Motor
G = Hollow Gimbal Motor
D = Direct Drive Motor

Stator Diameter (mm)

For example: 43 = 43 mm

Series-specific Parameter

J / S Series: Gear Ratio（For example:J4340: 40 = 40:1）
H / G / D Series: Stator Height (mm) (For example:G6220: 20 = 20 mm)

Bearing Type

(Blank) = Deep Groove Ball Bearing
P = Cross Roller Bearing
L = Lite Version； M:Max Version (L、M：not a bearing type)

Encoder Configuration

1E = Single Encoder
2E = Dual Encoders (with output-shaft single-turn absolute encoder)

Communication Interface

C = CAN
E = EtherCAT

Product Revision

① Default: V1.0 if omitted；② VX.X: X represents 0–9, denoting product revision versions

Rated Drive Voltage

48V = 48 V drive version(Only for 3507, 6220, and 43 series motors; not used for other wide-voltage motors. Default: 24 V if omitted.)

02 Special Naming Convention (JH Series：Harmonic Drive Motor)

DM - J H 11 - 51 - 2E C VX.X

(Brand) DAMIAO

Product Type

J = Joint Motor

Reducer Type

H = Harmonic Reducer

Size

Nominal pitch diameter reference: 11 ≈ 1.1 in

Gear Ratio

For example: 51 = 51:1

Encoder Configuration

2E = Dual Encoders (with output-shaft single-turn absolute encoder)

Communication Interface

C = CAN
E = EtherCAT

Product Revision

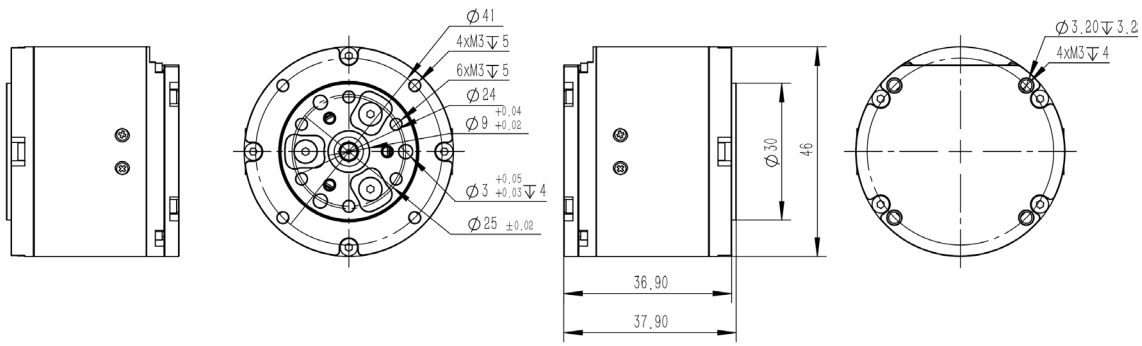
① Default: V1.0 if omitted；② VX.X: X represents 0–9, denoting product revision versions

Joint Motor DM-J3507-2EC

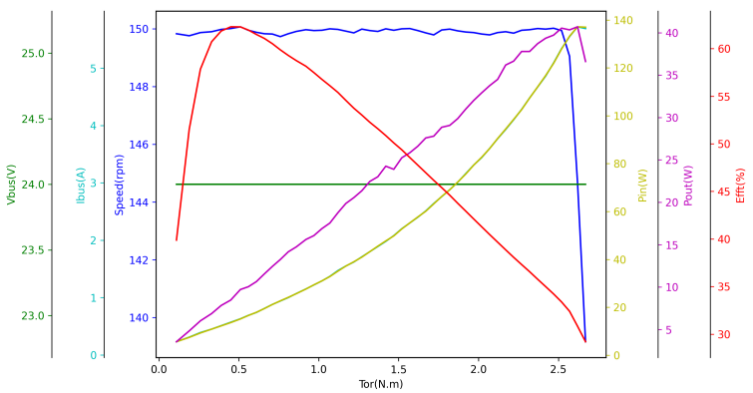


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	3
Peak Phase Current(A)	8.3
Rated Torque(Nm)	0.8
Peak Torque(Nm)	3
Rated Speed(rpm)	150
No-load Speed(rpm)	460
Rated Power(W)	12
Gear Ratio	7
Pair of Poles	14
Phase-to-Phase Inductance(uH)	235
Phase-to-Phase Resistance(Ω)	0.8
Motor Size Diameter*Height(mm)	46*37.9
Weight (g)	150
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

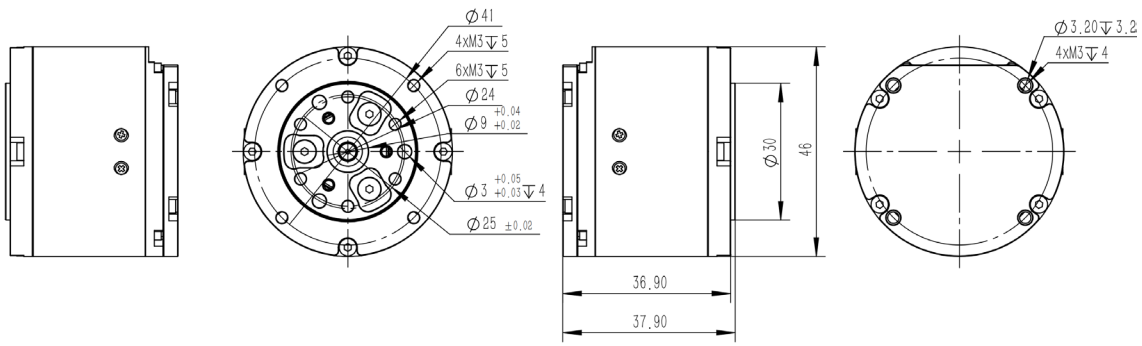


Joint Motor DM-J3507-2EC (48V)

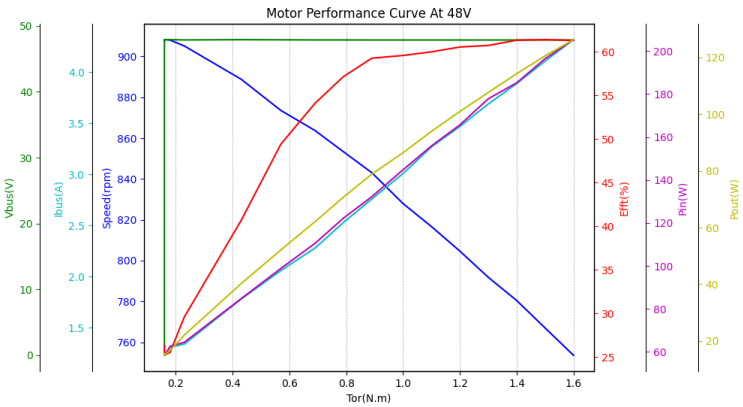


Specifications	
Supply Voltage(V)	48
Rated Phase Current(A)	3
Peak Phase Current(A)	8.3
Rated Torque(Nm)	0.8
Peak Torque(Nm)	3
Rated Speed(rpm)	150
No-load Speed(rpm)	900
Rated Power(W)	12
Gear Ratio	7
Pair of Poles	14
Phase-to-Phase Inductance(uH)	235
Phase-to-Phase Resistance(Ω)	0.8
Motor Size Diameter*Height(mm)	46*37.9
Weight (g)	150
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

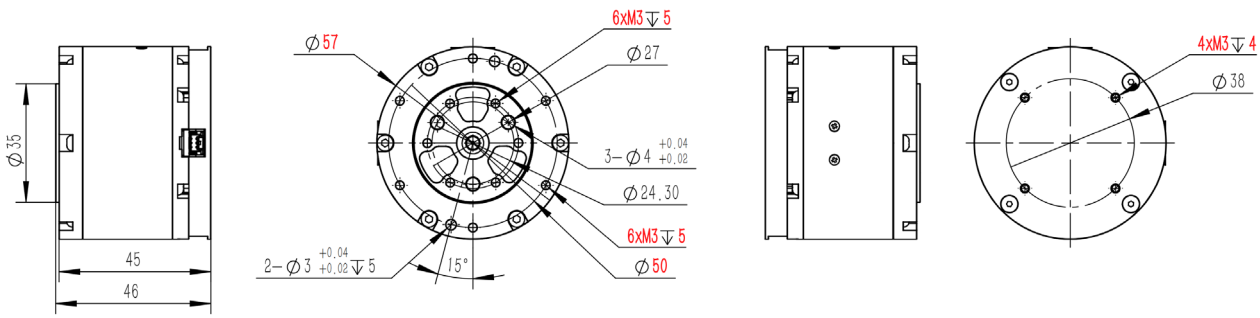


Joint Motor DM-J4310-2EC V1.1

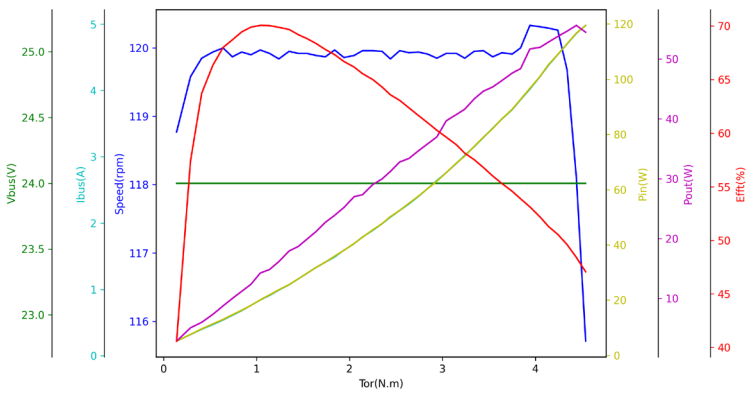


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	2.5
Peak Phase Current(A)	7.5
Rated Torque(Nm)	3
Peak Torque(Nm)	7
Rated Speed(rpm)	120
No-load Speed(rpm)	200
Rated Power(W)	37
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	345
Phase-to-Phase Resistance(Ω)	0.85
Motor Size Diameter*Height(mm)	57*46
Weight (g)	306
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

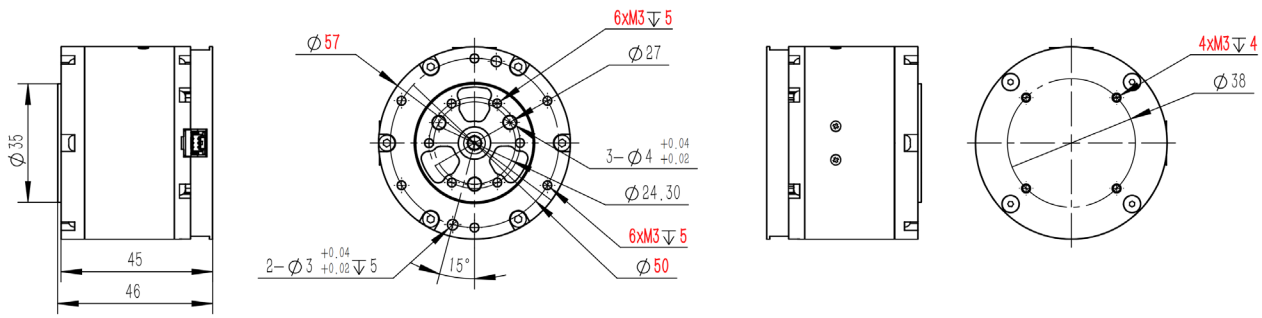


Joint Motor DM-J4310-2EC V1.1 (48V)

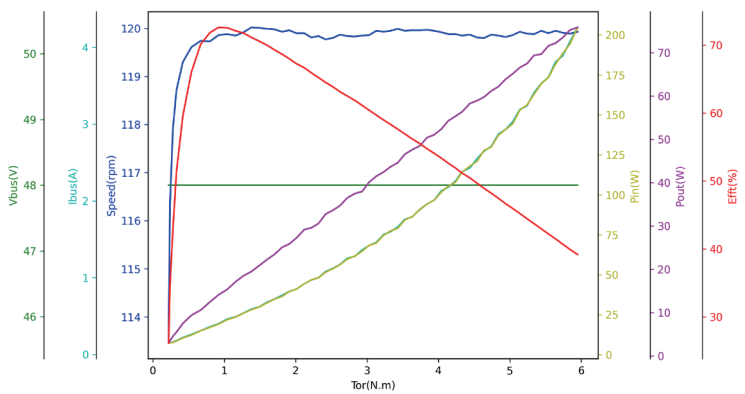


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	2.5
Peak Phase Current(A)	7.5
Rated Torque(Nm)	3
Peak Torque(Nm)	7
Rated Speed(rpm)	120
No-load Speed(rpm)	400
Rated Power(W)	37
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	345
Phase-to-Phase Resistance(Ω)	0.85
Motor Size Diameter*Height(mm)	57*46
Weight (g)	306
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

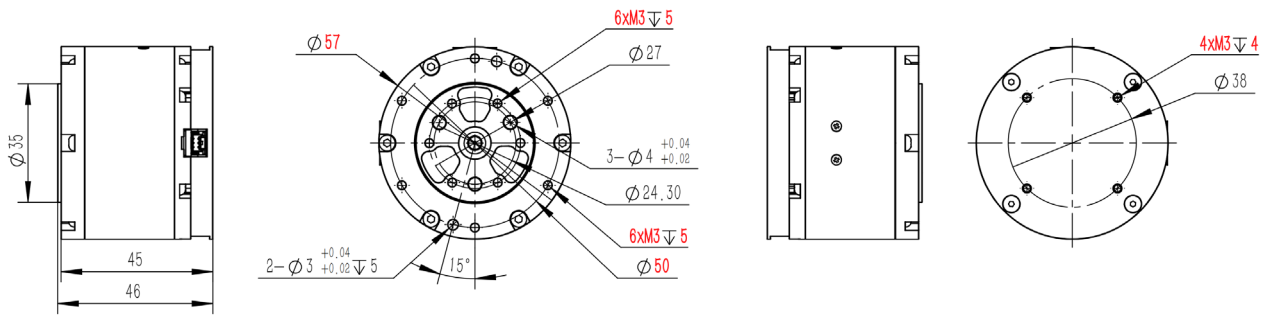


Joint Motor DM-J4310-2EC V1.2

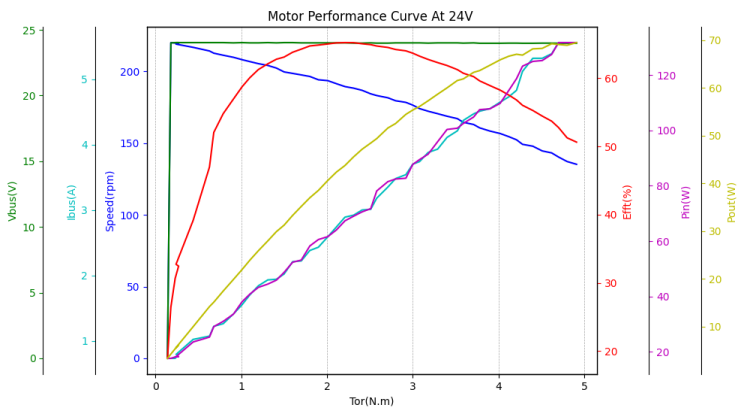


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	4.9
Peak Phase Current(A)	20
Rated Torque(Nm)	3.5
Peak Torque(Nm)	12.5
Rated Speed(rpm)	120
No-load Speed(rpm)	200
Rated Power(W)	44
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	320
Phase-to-Phase Resistance(Ω)	0.58
Motor Size Diameter*Height(mm)	57*46
Weight (g)	306
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

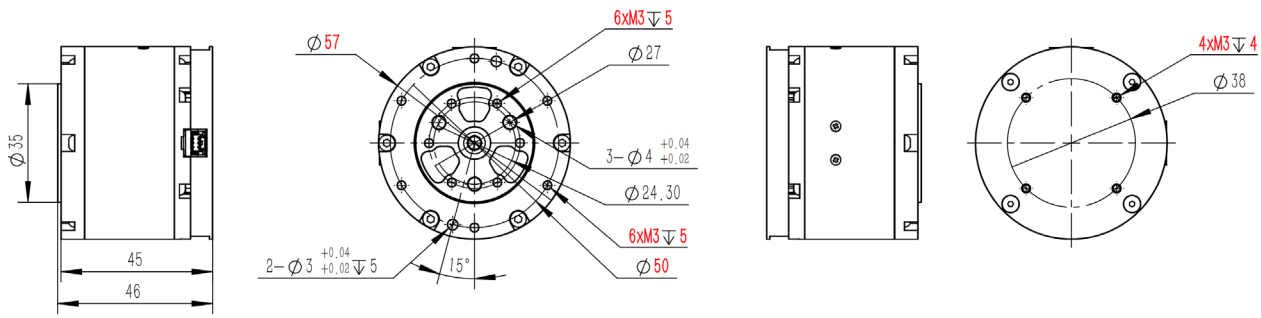


Joint Motor DM-J4310-2EC V1.2 (48V)

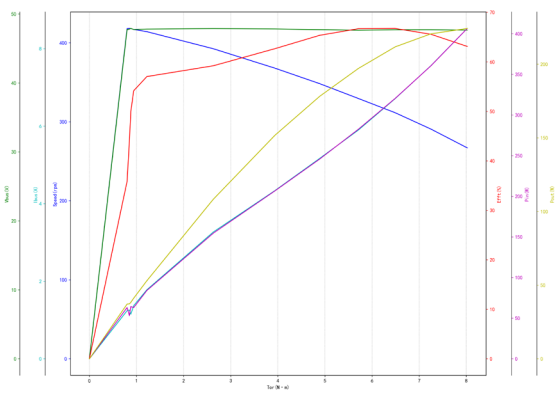


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	4.8
Peak Phase Current(A)	20
Rated Torque(Nm)	3.5
Peak Torque(Nm)	12.5
Rated Speed(rpm)	120
No-load Speed(rpm)	450
Rated Power(W)	44
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	320
Phase-to-Phase Resistance(Ω)	0.58
Motor Size Diameter*Height(mm)	57*46
Weight (g)	306
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

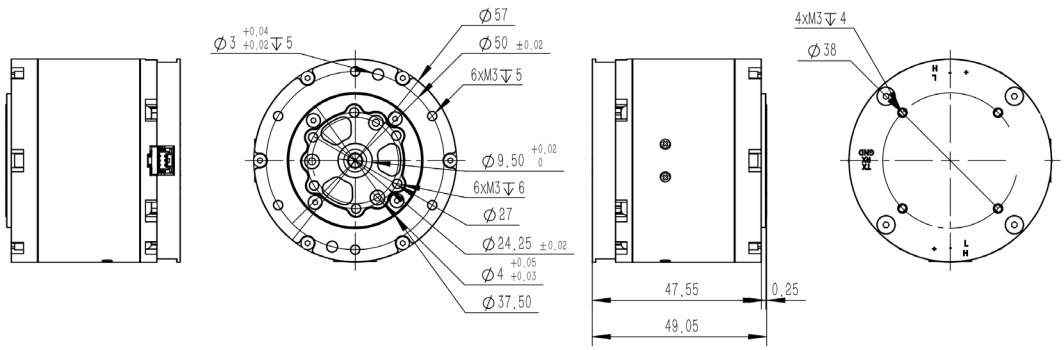


Joint Motor DM-J4310P-2EC

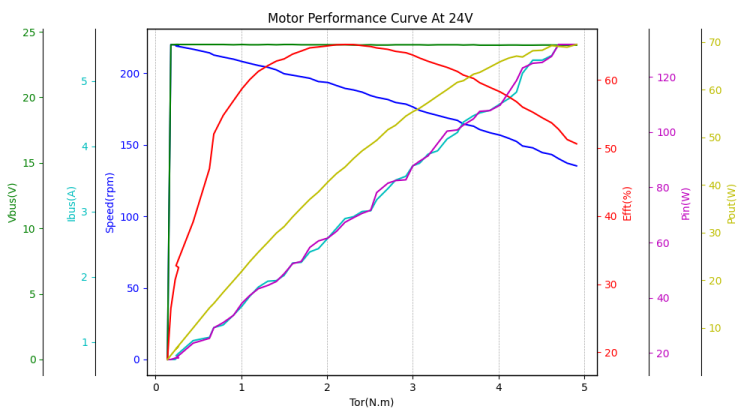


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	4.9
Peak Phase Current(A)	20
Rated Torque(Nm)	3.5
Peak Torque(Nm)	12.5
Rated Speed(rpm)	120
No-load Speed(rpm)	200
Rated Power(W)	43
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	320
Phase-to-Phase Resistance(Ω)	0.58
Motor Size Diameter*Height(mm)	57*49.05
Weight (g)	330
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

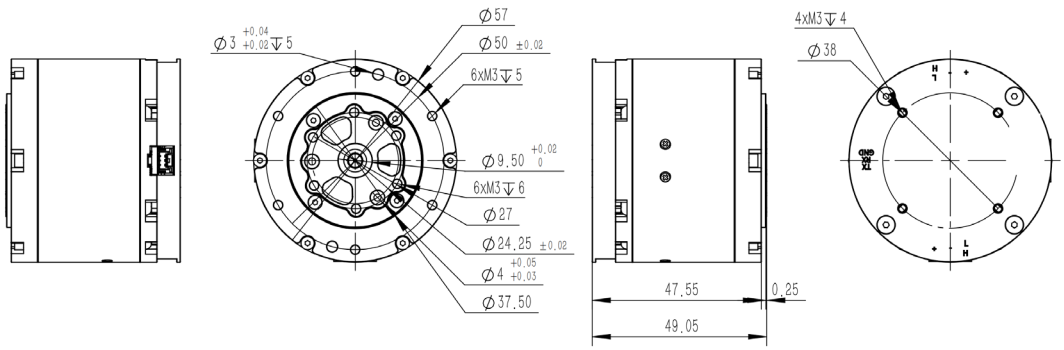


Joint Motor DM-J4310P-2EC（48V）

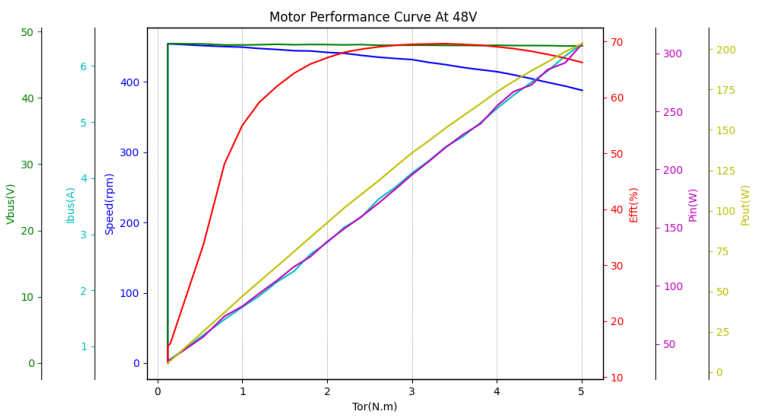


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	4.8
Peak Phase Current(A)	20
Rated Torque(Nm)	3.5
Peak Torque(Nm)	12.5
Rated Speed(rpm)	120
No-load Speed(rpm)	450
Rated Power(W)	43
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	320
Phase-to-Phase Resistance(Ω)	0.58
Motor Size Diameter*Height(mm)	57*49.05
Weight (g)	330
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

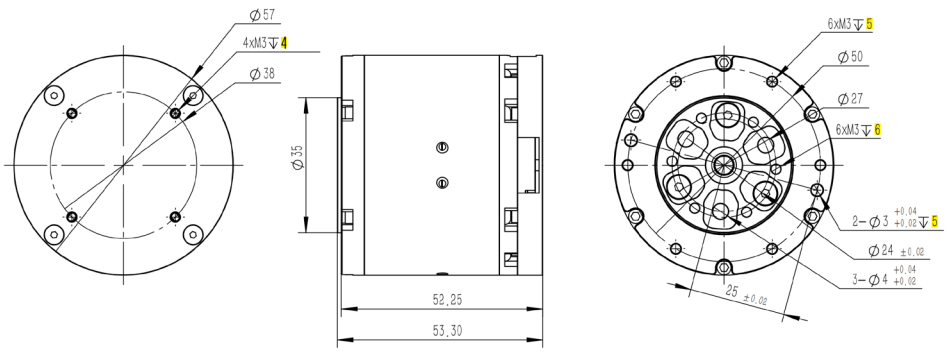


Joint Motor DM-J4340-2EC

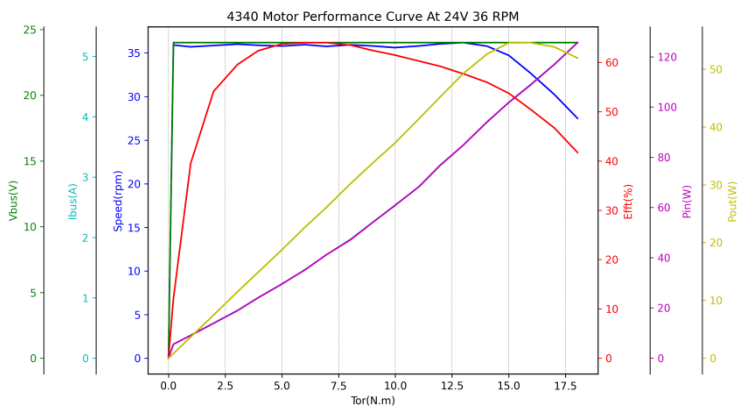


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	3
Peak Phase Current(A)	8
Rated Torque(Nm)	9
Peak Torque(Nm)	27
Rated Speed(rpm)	36
No-load Speed(rpm)	52.5
Rated Power(W)	33
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	360
Phase-to-Phase Resistance(Ω)	0.88
Motor Size Diameter*Height(mm)	57*53.3
Weight (g)	362
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

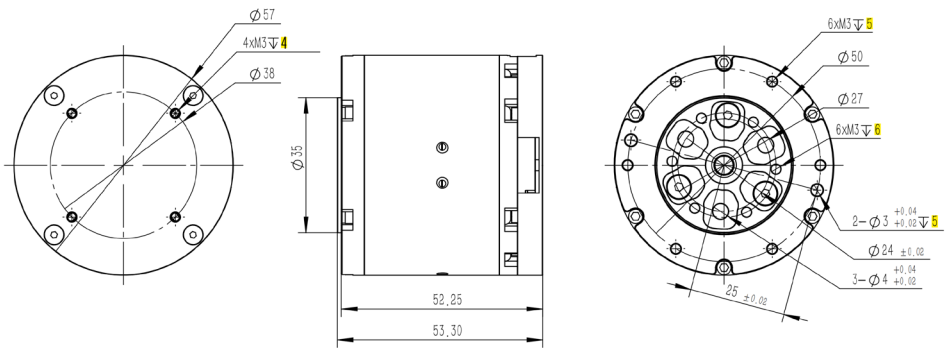


Joint Motor DM-J4340-2EC（48V）

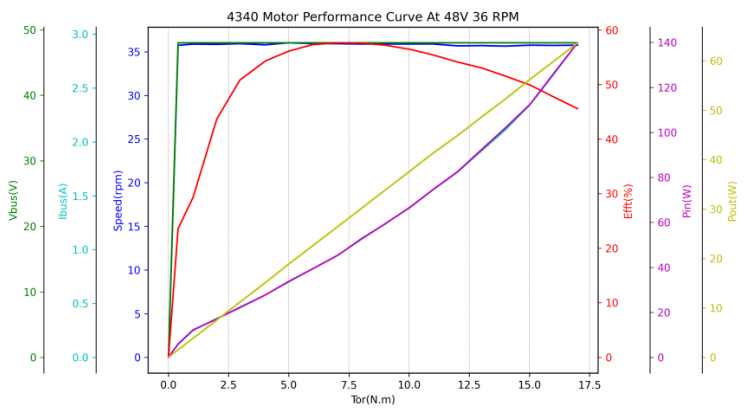


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	2.5
Peak Phase Current(A)	9
Rated Torque(Nm)	9
Peak Torque(Nm)	27
Rated Speed(rpm)	36
No-load Speed(rpm)	100
Rated Power(W)	33
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	360
Phase-to-Phase Resistance(Ω)	0.88
Motor Size Diameter*Height(mm)	57*53.3
Weight (g)	362
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

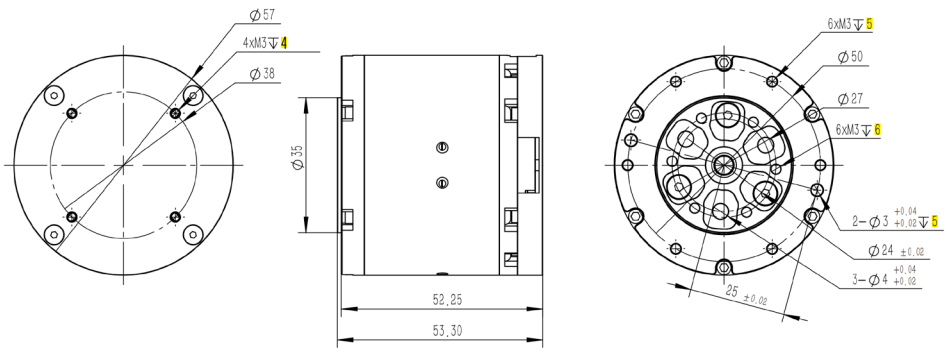


Joint Motor DM-J4340-2EC V1.1

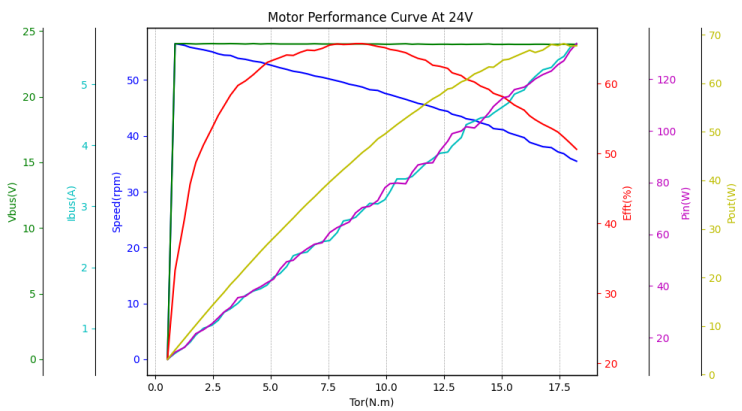


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	4
Peak Phase Current(A)	19
Rated Torque(Nm)	12
Peak Torque(Nm)	40
Rated Speed(rpm)	36
No-load Speed(rpm)	56
Rated Power(W)	45
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	290
Phase-to-Phase Resistance(Ω)	0.72
Motor Size Diameter*Height(mm)	57*53.3
Weight (g)	362
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

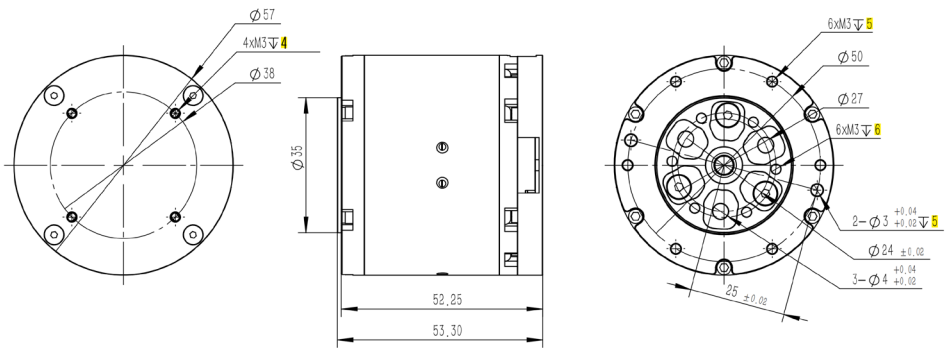


Joint Motor DM-J4340-2EC V1.1（48V）

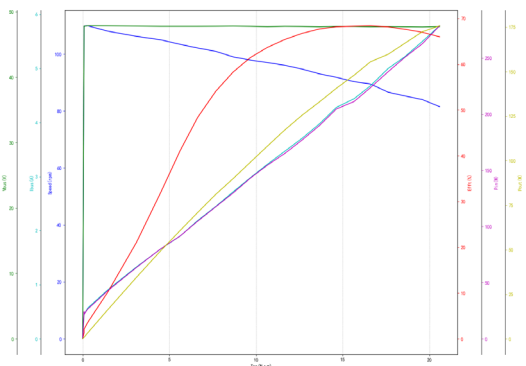


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	4
Peak Phase Current(A)	19
Rated Torque(Nm)	12
Peak Torque(Nm)	40
Rated Speed(rpm)	36
No-load Speed(rpm)	112
Rated Power(W)	45
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	290
Phase-to-Phase Resistance(Ω)	0.72
Motor Size Diameter*Height(mm)	57*53.3
Weight (g)	362
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

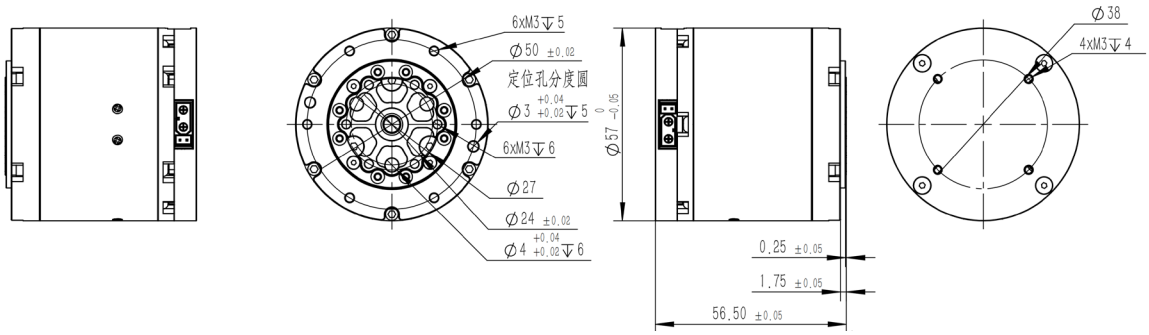


Joint Motor DM-J4340P-2EC

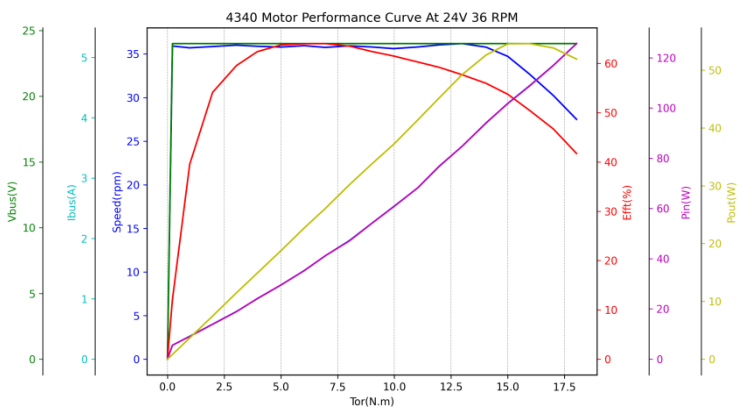


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	3
Peak Phase Current(A)	8
Rated Torque(Nm)	9
Peak Torque(Nm)	27
Rated Speed(rpm)	36
No-load Speed(rpm)	52.5
Rated Power(W)	33
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	360
Phase-to-Phase Resistance(Ω)	0.88
Motor Size Diameter*Height(mm)	57*56.5
Weight (g)	381
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

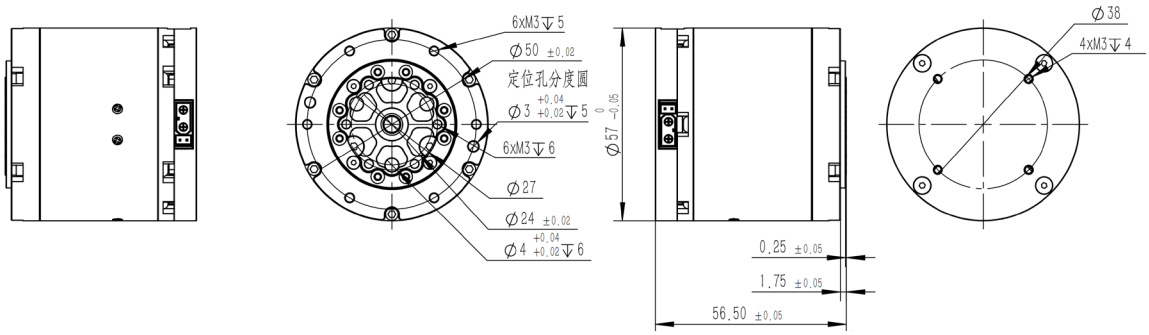


Joint Motor DM-J4340P-2EC（48V）

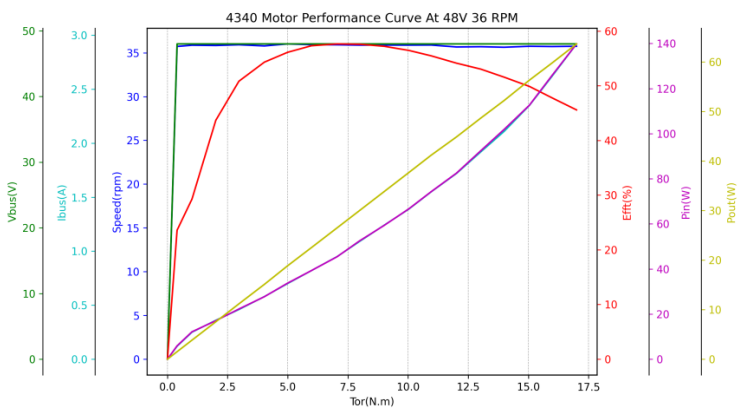


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	2.5
Peak Phase Current(A)	9
Rated Torque(Nm)	9
Peak Torque(Nm)	27
Rated Speed(rpm)	36
No-load Speed(rpm)	100
Rated Power(W)	33
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	360
Phase-to-Phase Resistance(Ω)	0.88
Motor Size Diameter*Height(mm)	57*56.5
Weight (g)	381
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

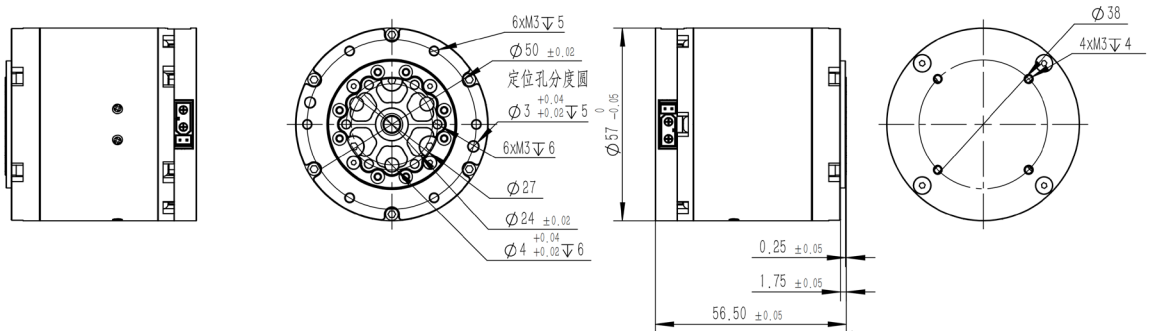


Joint Motor DM-J4340P-2EC V1.1

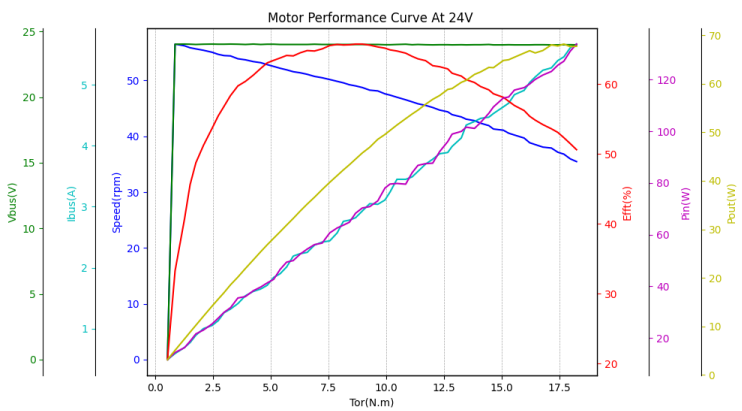


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	4
Peak Phase Current(A)	19
Rated Torque(Nm)	12
Peak Torque(Nm)	40
Rated Speed(rpm)	36
No-load Speed(rpm)	56
Rated Power(W)	45
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	290
Phase-to-Phase Resistance(Ω)	0.72
Motor Size Diameter*Height(mm)	57*56.5
Weight (g)	381
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

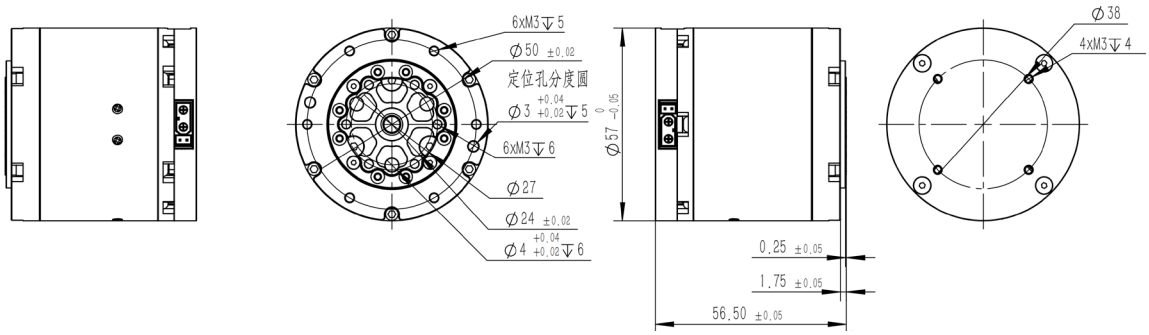


Joint Motor DM-J4340P-2EC V1.1（48V）

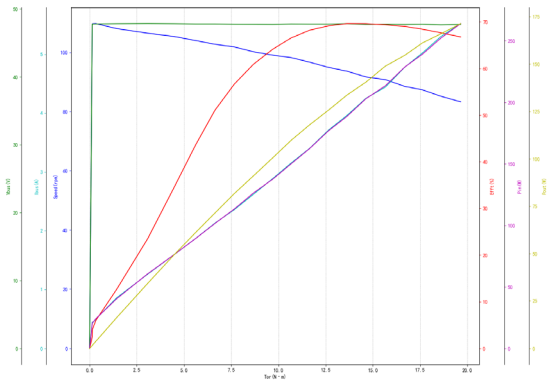


Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	4
Peak Phase Current(A)	19
Rated Torque(Nm)	12
Peak Torque(Nm)	40
Rated Speed(rpm)	36
No-load Speed(rpm)	112
Rated Power(W)	45
Gear Ratio	40
Pair of Poles	14
Phase-to-Phase Inductance(uH)	290
Phase-to-Phase Resistance(Ω)	0.72
Motor Size Diameter*Height(mm)	57*56.5
Weight (g)	381
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

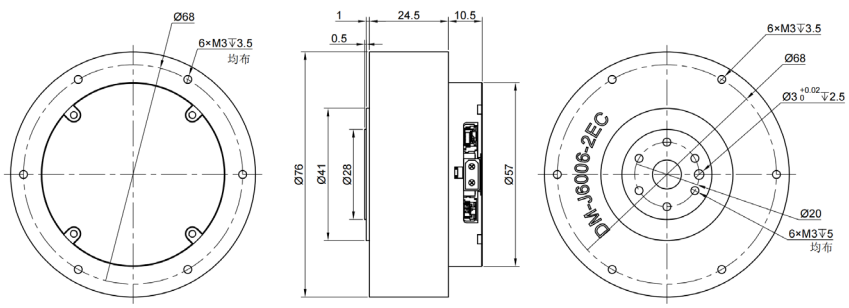


Joint Motor DM-J6006-2EC

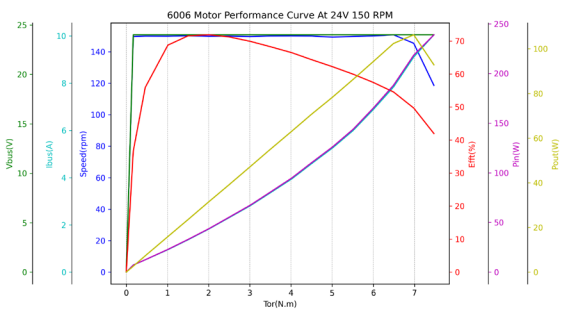


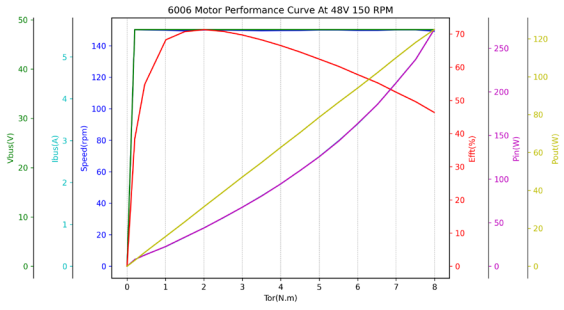
Specifications	
Supply Voltage(V)	24V（Support 24-48V）
Rated Phase Current(A)	5.96
Peak Phase Current(A)	17.6
Rated Torque(Nm)	4
Peak Torque(Nm)	11
Rated Speed(rpm)	150
No-load Speed(rpm)	226@24V 408@48V
Rated Power(W)	62
Gear Ratio	6
Pair of Poles	14
Phase-to-Phase Inductance(uH)	270
Phase-to-Phase Resistance(Ω)	0.435
Motor Size Diameter*Height(mm)	76*36.5
Weight (g)	335
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



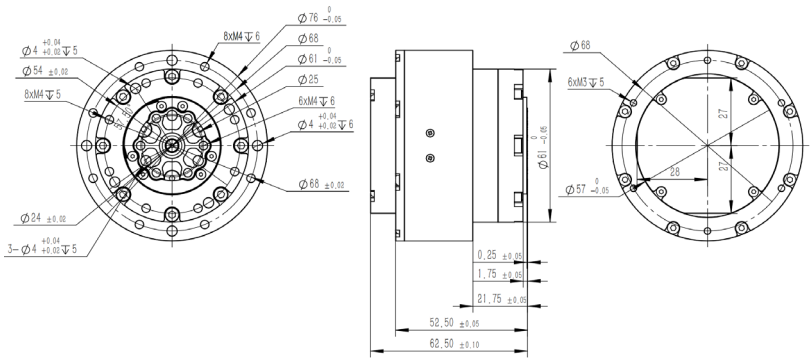


Joint Motor DM-J6248P-2EC

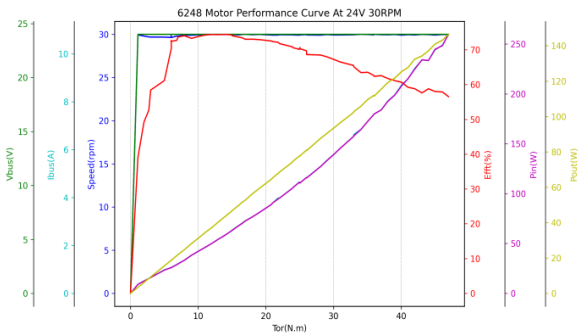


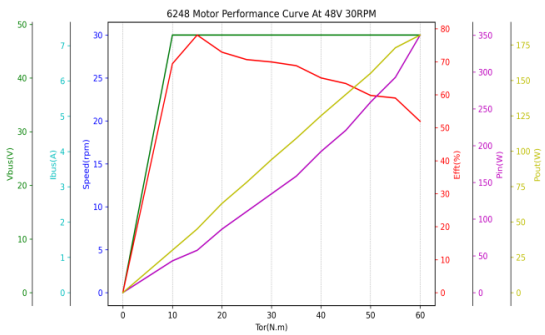
Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	11.3A/7A@24V 11.3A/3.7A@48V
Peak Phase Current(A)	38.7A/16A@24V 38.7A/8A@48V
Rated Torque(Nm)	30
Peak Torque(Nm)	97
Rated Speed(rpm)	40
No-load Speed(rpm)	60@24V；115@48V
Rated Power(W)	125
Gear Ratio	48
Pair of Poles	14
Phase-to-Phase Inductance(uH)	85
Phase-to-Phase Resistance(Ω)	0.17
Motor Size Diameter*Height(mm)	76*62.5
Weight (g)	660
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



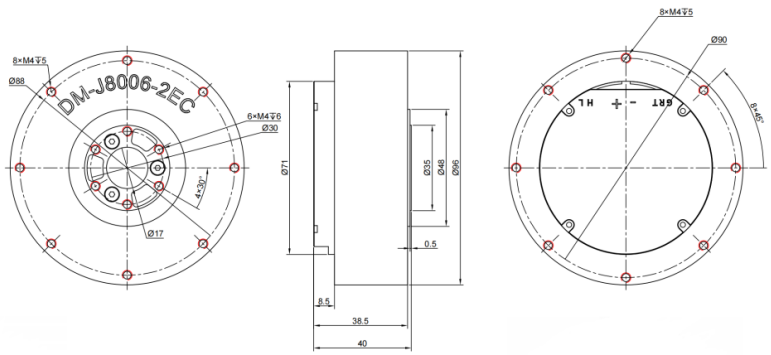


Joint Motor DM-J8006-2EC V1.1

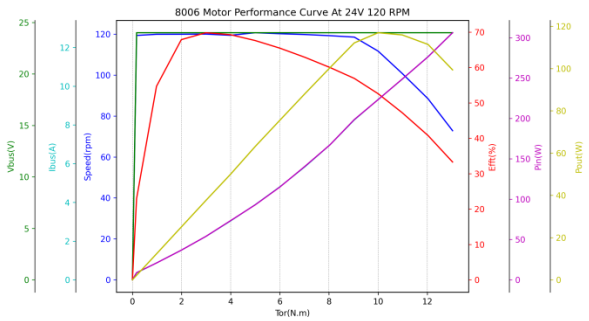


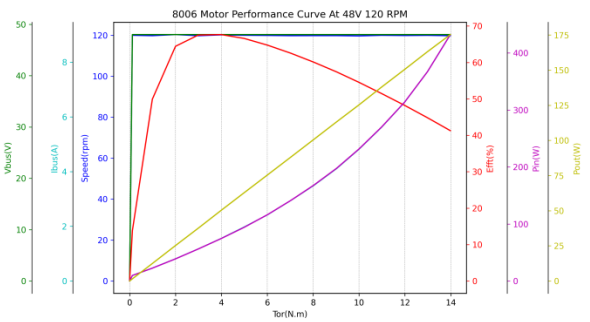
Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	9
Peak Phase Current(A)	21
Rated Torque(Nm)	8
Peak Torque(Nm)	20
Rated Speed(rpm)	120
No-load Speed(rpm)	194.2@24V; 392.8@48V
Rated Power(W)	100
Gear Ratio	6
Pair of Poles	21
Phase-to-Phase Inductance(uH)	215
Phase-to-Phase Resistance(Ω)	0.4
Motor Size Diameter*Height(mm)	96*40
Weight (g)	559
Encoder	Magnetic encoder /Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



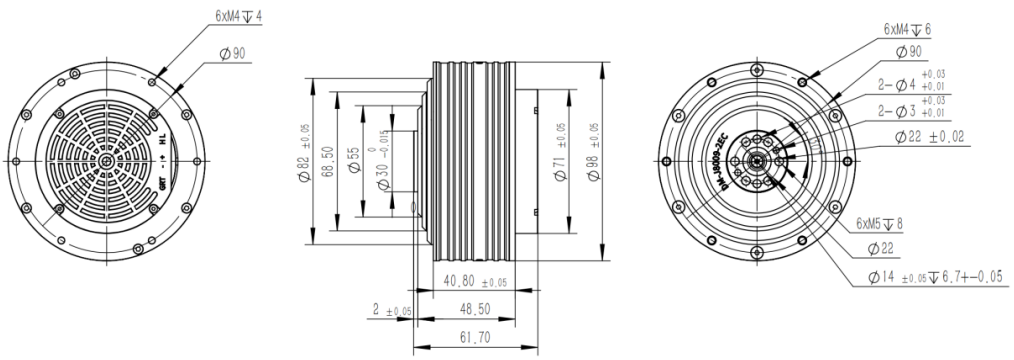


Joint Motor DM-J8009-2EC

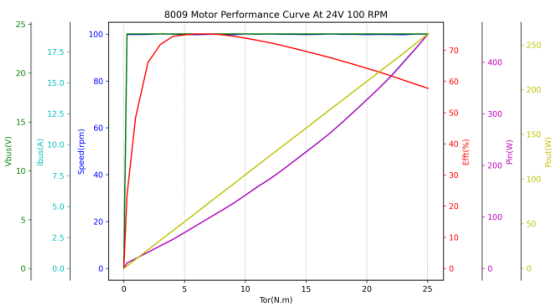


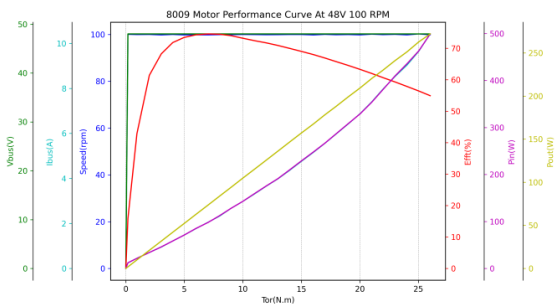
Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	18.6
Peak Phase Current(A)	40
Rated Torque(Nm)	20
Peak Torque(Nm)	40
Rated Speed(rpm)	100@24V ; 200@48V
No-load Speed(rpm)	168@24V ; 335@48V
Rated Power(W)	209@24V ; 418@48V
Gear Ratio	9
Pair of Poles	21
Phase-to-Phase Inductance(uH)	80
Phase-to-Phase Resistance(Ω)	0.145
Motor Size Diameter*Height(mm)	98*61.7
Weight (g)	896
Encoder	Magnetic encoder /Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



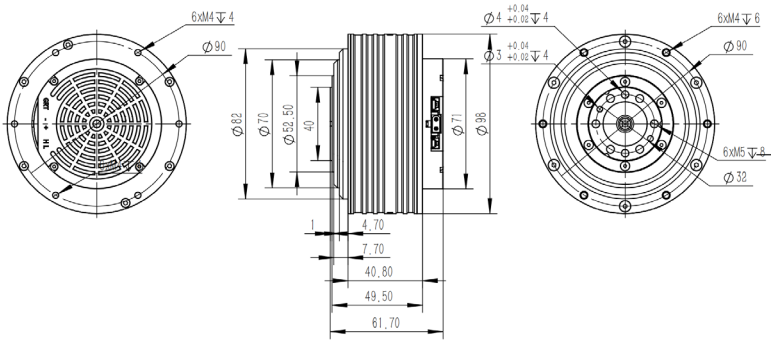


Joint Motor DM-J8009P-2EC

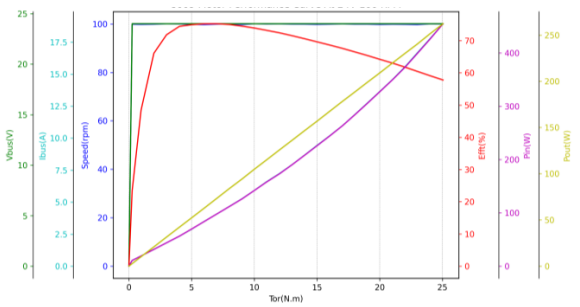


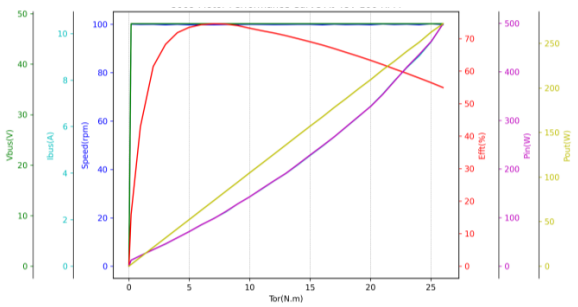
Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	18.6
Peak Phase Current(A)	40
Rated Torque(Nm)	20
Peak Torque(Nm)	40
Rated Speed(rpm)	100@24V; 200@48V
No-load Speed(rpm)	168@24V ; 335@48V
Rated Power(W)	209@24V ; 418@48V
Gear Ratio	9
Pair of Poles	21
Phase-to-Phase Inductance(uH)	80
Phase-to-Phase Resistance(Ω)	0.145
Motor Size Diameter*Height(mm)	98*61.7
Weight (g)	963
Encoder	Magnetic encoder/Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



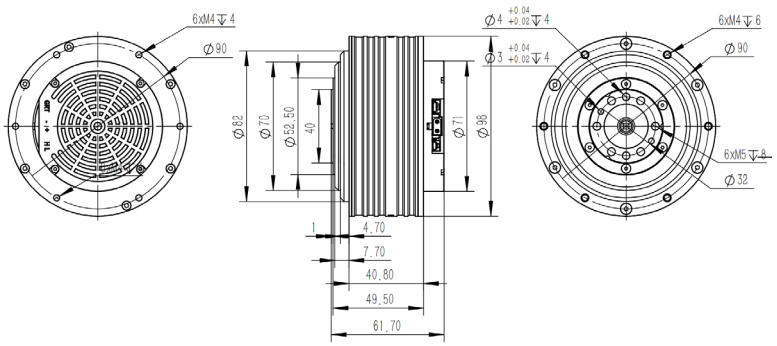


Joint Motor DM-J8009P-2EC V1.1

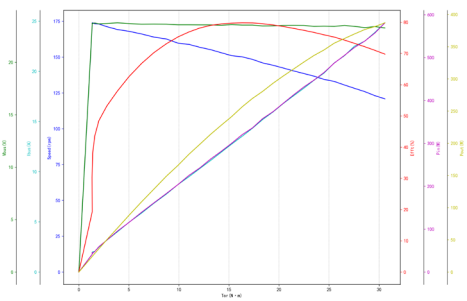


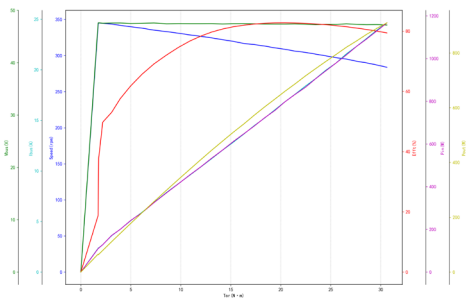
Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	18.6
Peak Phase Current(A)	40
Rated Torque(Nm)	20
Peak Torque(Nm)	40
Rated Speed(rpm)	100@24V; 200@48V
No-load Speed(rpm)	168@24V ; 335@48V
Rated Power(W)	209@24V ; 418@48V
Gear Ratio	9
Pair of Poles	21
Phase-to-Phase Inductance(uH)	80
Phase-to-Phase Resistance(Ω)	0.145
Motor Size Diameter*Height(mm)	98*61.7
Weight (g)	950
Encoder	Magnetic encoder/Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



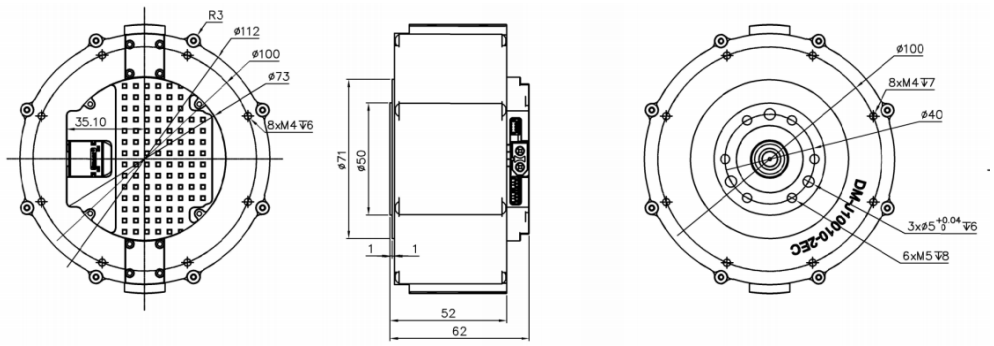


Joint Motor DM-J10010-2EC

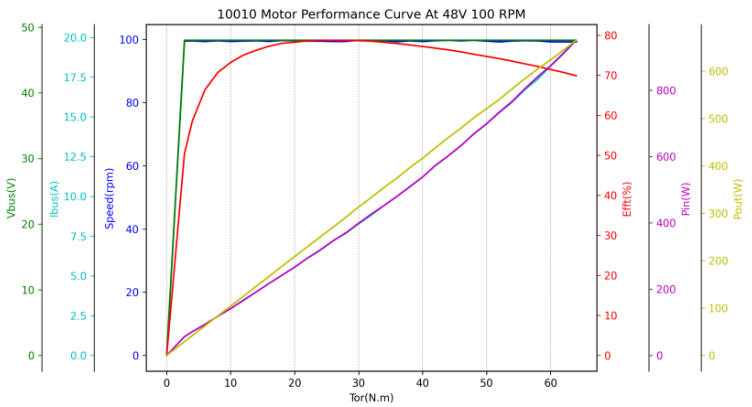


Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	18.4
Peak Phase Current(A)	80.2
Rated Torque(Nm)	40
Peak Torque(Nm)	150
Rated Speed(rpm)	50@24V ; 100@48V
No-load Speed(rpm)	75@24V ; 150@48V
Rated Power(W)	209@24V ; 418@48V
Gear Ratio	10
Pair of Poles	21
Phase-to-Phase Inductance(uH)	100
Phase-to-Phase Resistance(Ω)	0.125
Motor Size Diameter*Height(mm)	112*62 (Maximum OuterDimensions:120)
Weight (g)	1485
Encoder	Magnetic encoder /Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

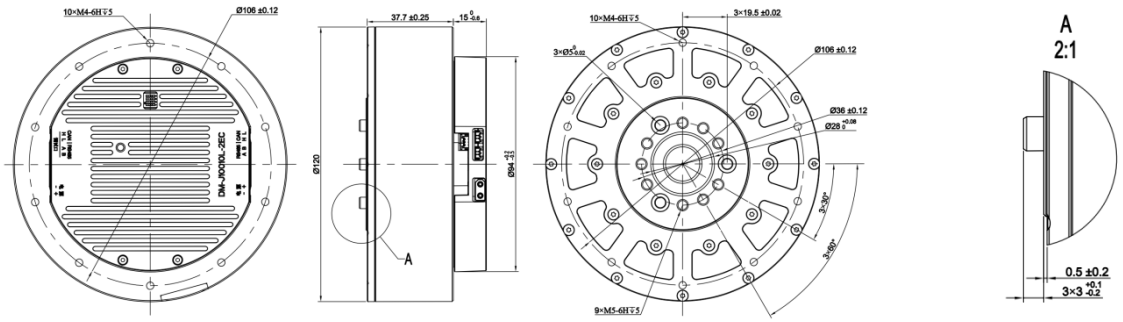


Joint Motor DM-J10010L-2EC

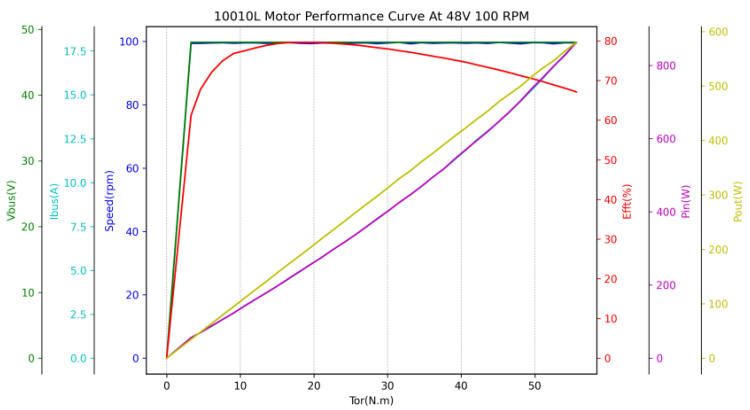


Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	23.5
Peak Phase Current(A)	95
Rated Torque(Nm)	40
Peak Torque(Nm)	120
Rated Speed(rpm)	70@24V ; 100@48V
No-load Speed(rpm)	100@24V; 200@48V
Rated Power(W)	293@24V ; 418@48V
Gear Ratio	10
Pair of Poles	21
Phase-to-Phase Inductance(uH)	85
Phase-to-Phase Resistance(Ω)	0.11
Motor Size Diameter*Height(mm)	120*53 (Height excluding locating pins)
Weight (g)	1375
Encoder	Magnetic encoder/ Dual encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

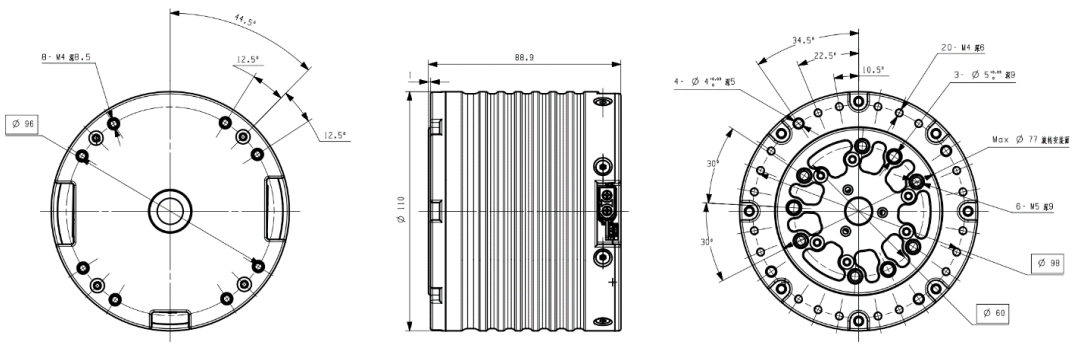


Joint Motor DM-J10422P-2EC

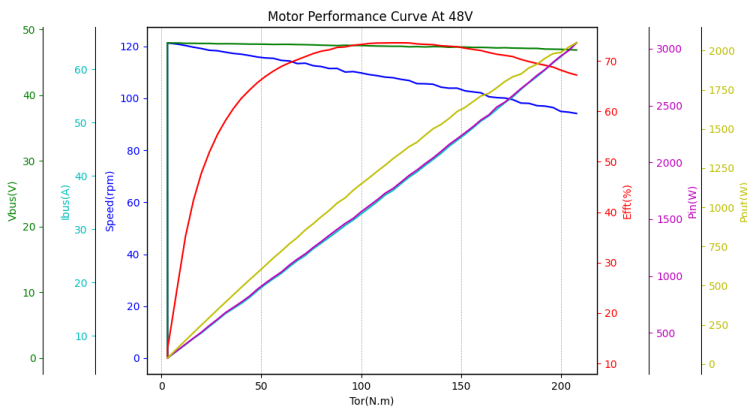


Specifications	
Supply Voltage(V)	48
Rated Phase Current(A)	44.5
Peak Phase Current(A)	165
Rated Torque(Nm)	100
Peak Torque(Nm)	400
Rated Speed(rpm)	100
No-load Speed(rpm)	120
Rated Power(W)	1047
Gear Ratio	22
Pair of Poles	16
Phase-to-Phase Inductance(uH)	38
Phase-to-Phase Resistance(Ω)	0.081
Motor Size Diameter*Height(mm)	110*89.9
Weight (g)	2700
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

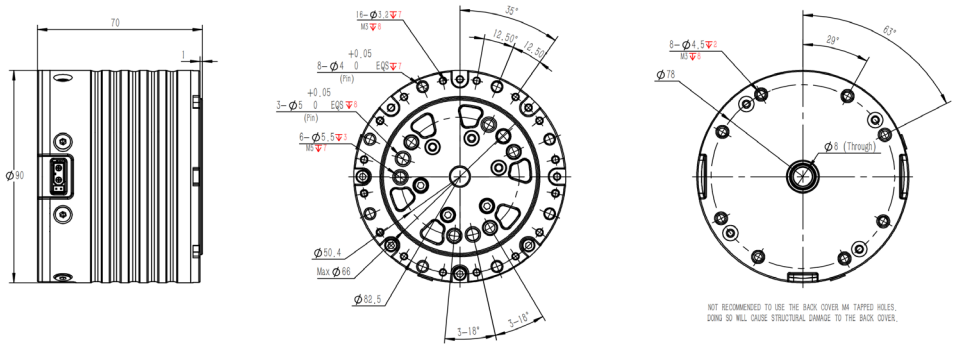


Joint Motor DM-J8520P-2EC

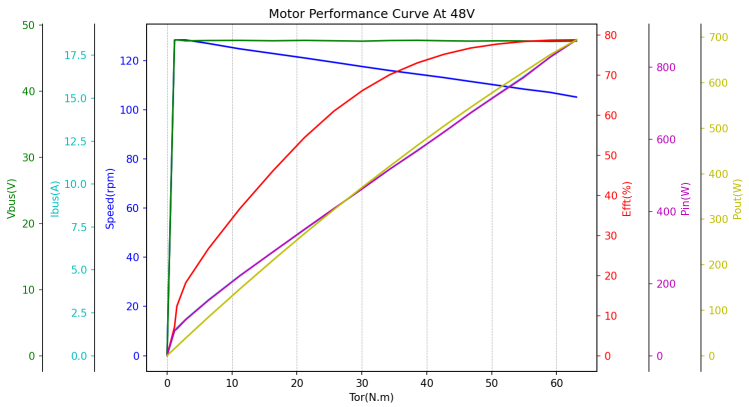


Specifications	
Supply Voltage(V)	48
Rated Phase Current(A)	13.86
Peak Phase Current(A)	41.6
Rated Torque(Nm)	50
Peak Torque(Nm)	176
Rated Speed(rpm)	120
No-load Speed(rpm)	124
Rated Power(W)	628
Gear Ratio	20
Pair of Poles	16
Phase-to-Phase Inductance(uH)	75
Phase-to-Phase Resistance(Ω)	0.1
Motor Size Diameter*Height(mm)	90*70
Weight (g)	1392
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

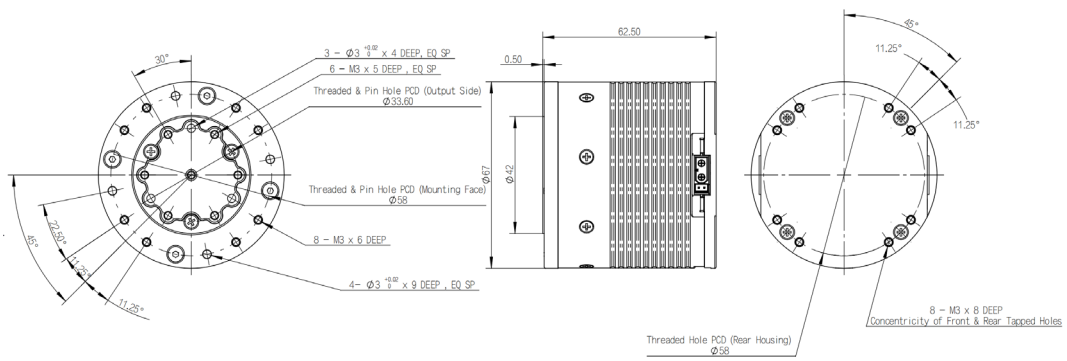


Joint Motor DM-J6216L-2EC

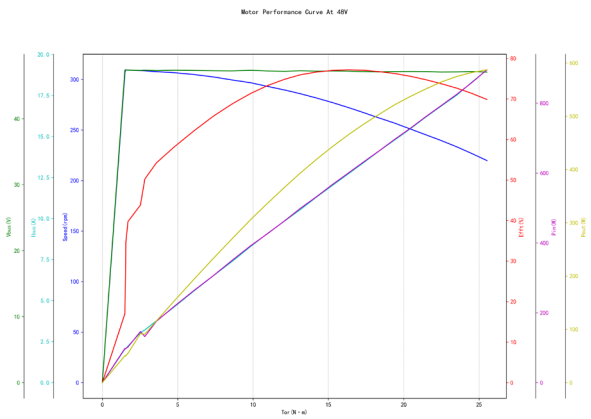


Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	7.48@24V；7.74@48V
Peak Phase Current(A)	39.5@24V；27@48V
Rated Torque(Nm)	10
Peak Torque(Nm)	45
Rated Speed(rpm)	200
No-load Speed(rpm)	260
Rated Power(W)	209
Gear Ratio	16
Pair of Poles	10
Phase-to-Phase Inductance(uH)	184
Phase-to-Phase Resistance(Ω)	0.16
Motor Size Diameter*Height(mm)	67*62.5
Weight (g)	740
Encoder	Inductive + Magnetic Encoder Dual Encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

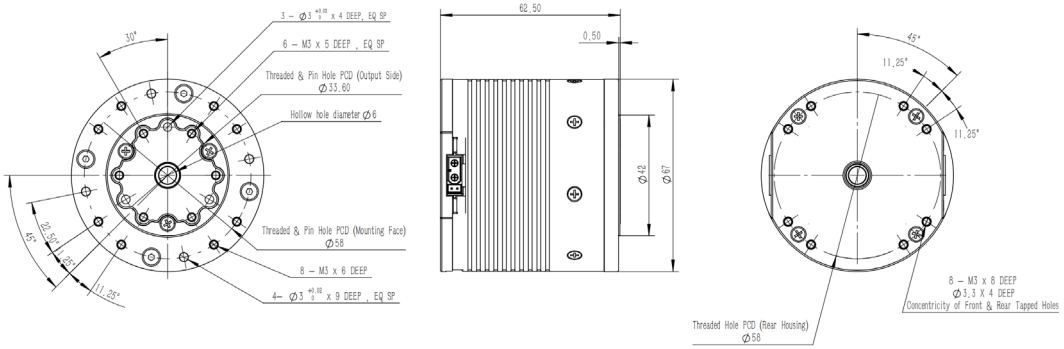


Joint Motor DM-J6216P-2EC

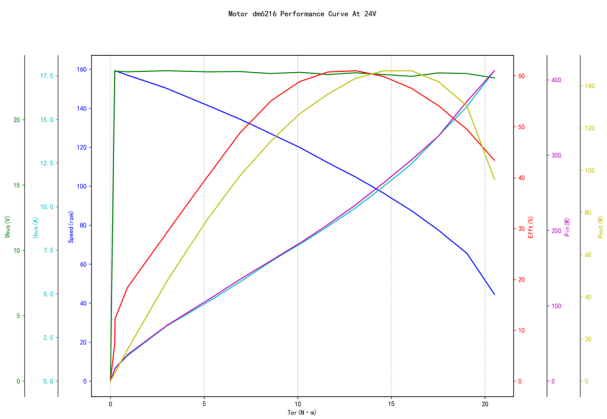


Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	20.8@24V；22@48V
Peak Phase Current(A)	36@24V；70@48V
Rated Torque(Nm)	10
Peak Torque(Nm)	45
Rated Speed(rpm)	200
No-load Speed(rpm)	260
Rated Power(W)	209
Gear Ratio	16
Pair of Poles	10
Phase-to-Phase Inductance(uH)	184
Phase-to-Phase Resistance(Ω)	0.16
Motor Size Diameter*Height(mm)	67*62.5
Weight (g)	740
Encoder	Inductive + Magnetic Encoder Dual Encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve

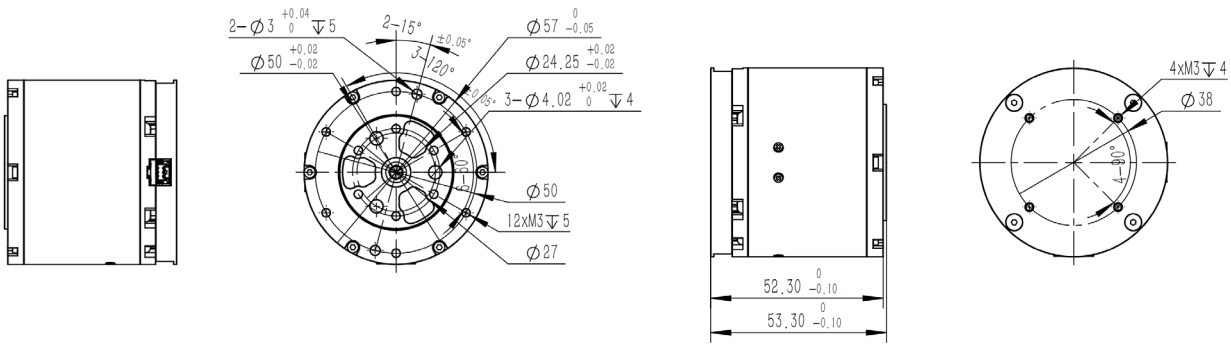


Joint Motor DM-J4310M-2EC

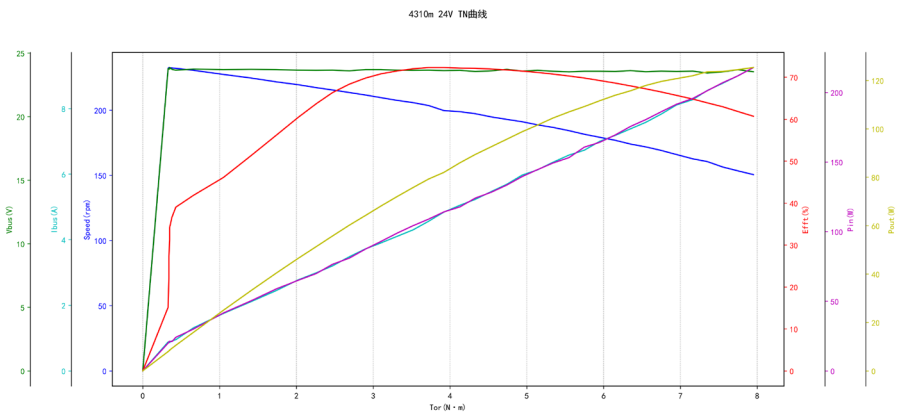


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	8.49
Peak Phase Current(A)	32
Rated Torque(Nm)	5.2
Peak Torque(Nm)	18
Rated Speed(rpm)	120
No-load Speed(rpm)	232
Rated Power(W)	65.3
Gear Ratio	10
Pair of Poles	14
Phase-to-Phase Inductance(uH)	210.6415
Phase-to-Phase Resistance(Ω)	0.327465
Motor Size Diameter*Height(mm)	57*53.3
Weight (g)	370
Encoder	Magnetic Encoder Dual Encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



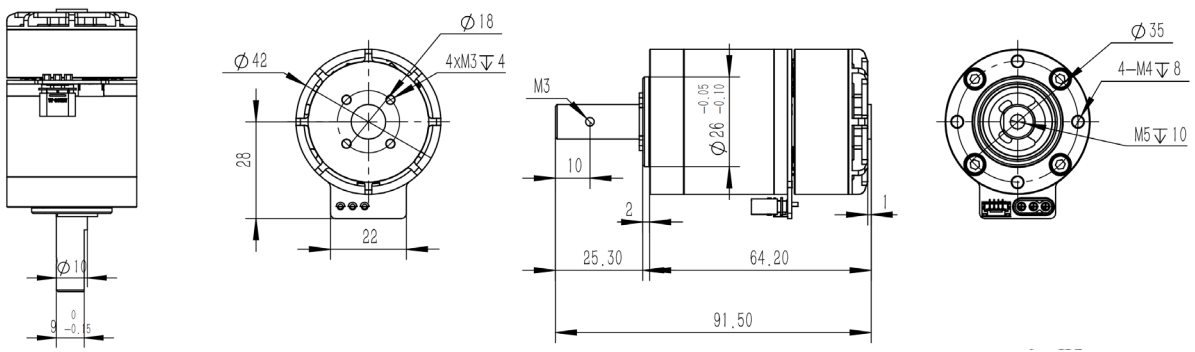
Separable Motor DM-S3519-1EC



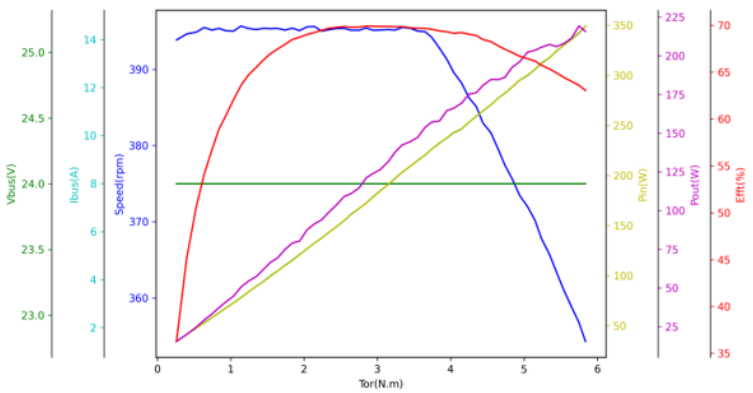


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	9.2
Peak Phase Current(A)	20.5
Rated Torque(Nm)	3.5
Peak Torque(Nm)	7.8
Rated Speed(rpm)	395
No-load Speed(rpm)	435
Rated Power(W)	144
Gear Ratio	19.203
Pair of Poles	7
Phase-to-Phase Inductance(uH)	55
Phase-to-Phase Resistance(Ω)	0.2
Motor Size Diameter*Height(mm)	42*91.5
Weight (g)	396
Encoder	Hall Encoder Incremental Encoder
Communication Interface	CAN、CANFD

Dimension



Performance Curve



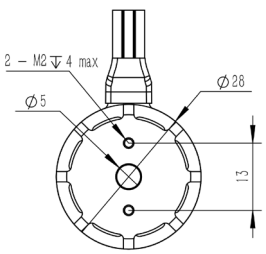
Separable Motor DM-S2325-1EC

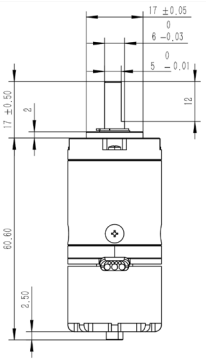


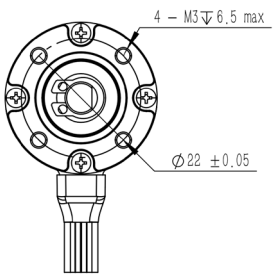


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	5
Peak Phase Current(A)	13
Rated Torque(Nm)	1.35
Peak Torque(Nm)	5
Rated Speed(rpm)	380
No-load Speed(rpm)	560
Rated Power(W)	53
Gear Ratio	25
Pair of Poles	7
Phase-to-Phase Inductance(uH)	75
Phase-to-Phase Resistance(Ω)	0.292
Motor Size Diameter*Height(mm)	28*77.6
Weight (g)	172
Encoder	Hall Encoder Incremental Encoder
Communication Interface	CAN、CANFD

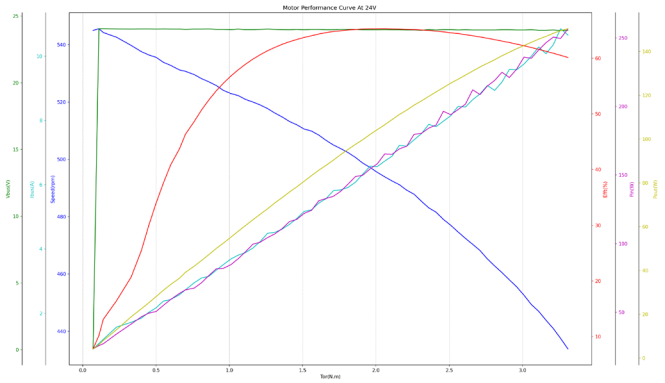
Dimension







Performance Curve

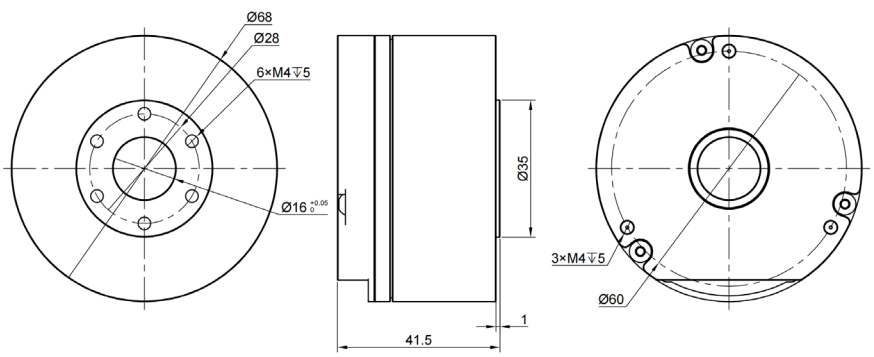


Hollow Gimbal DM-G6220

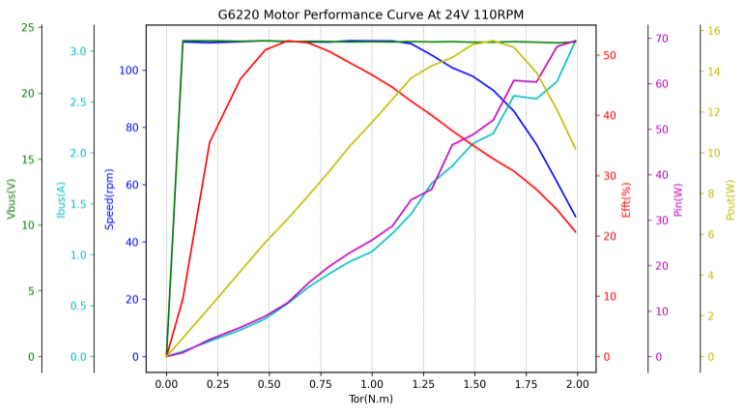


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	2.3
Peak Phase Current(A)	4.6
Rated Torque(Nm)	1.3
Peak Torque(Nm)	2.7
Rated Speed(rpm)	110
No-load Speed(rpm)	300
Rated Power(W)	15
Gear Ratio	1
Pair of Poles	14
Phase-to-Phase Inductance(uH)	3000
Phase-to-Phase Resistance(Ω)	3
Motor Size Diameter*Height(mm)	68*41.5
Weight (g)	493.5
Encoder	Magnetic Encoder Absolute
Communication Interface	CAN、CANFD

Dimension



Performance Curve

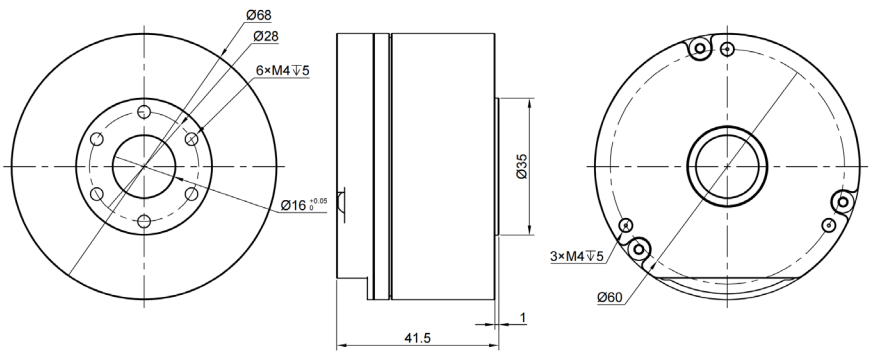


Hollow Gimbal DM-G6220 （48V）

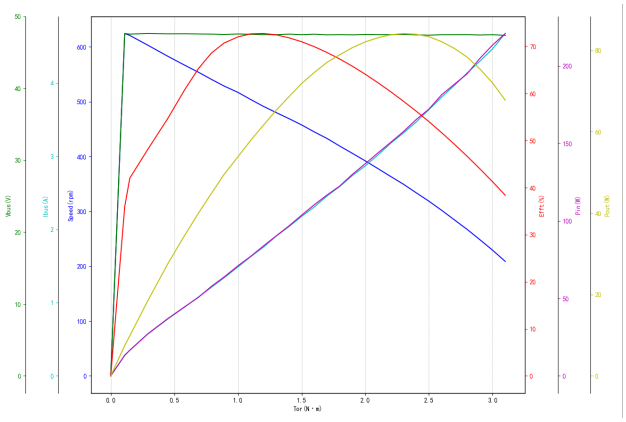


Specifications	
Supply Voltage(V)	48
Rated Phase Current(A)	0.8
Peak Phase Current(A)	7.82
Rated Torque(Nm)	1.3
Peak Torque(Nm)	4.9
Rated Speed(rpm)	110
No-load Speed(rpm)	650
Rated Power(W)	15
Gear Ratio	1
Pair of Poles	14
Phase-to-Phase Inductance(uH)	2376
Phase-to-Phase Resistance(Ω)	2.84
Motor Size Diameter*Height(mm)	68*41.5
Weight (g)	493.5
Encoder	Magnetic Encoder Absolute
Communication Interface	CAN、CANFD

Dimension



Performance Curve

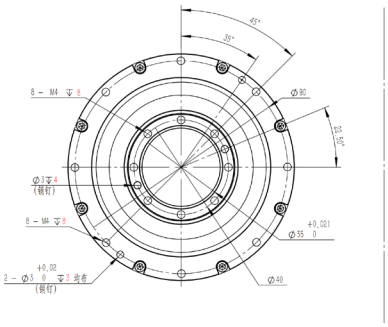


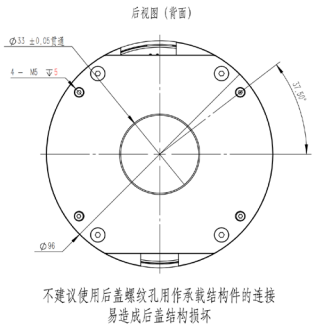
Hollow Gimbal DM-G8518

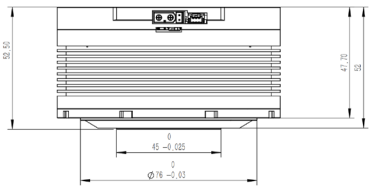


Specifications	
Supply Voltage(V)	24V（Support 24-48V）
Rated Phase Current(A)	9.55A/10.51A@24V 9.55A/10.05A@48V
Peak Phase Current(A)	36.1A/33.05A@24V 36.1A/18.05A@48V
Rated Torque(Nm)	1.9
Peak Torque(Nm)	8.78
Rated Speed(rpm)	1000@24V；2000@48V
No-load Speed(rpm)	1290@24V；2590@48V
Rated Power(W)	199@24V；397@48V
Gear Ratio	1
Pair of Poles	16
Phase-to-Phase Inductance(uH)	105
Phase-to-Phase Resistance(Ω)	0.225
Motor Size Diameter*Height(mm)	96*52.5
Weight (g)	831
Encoder	Inductance Absolute
Communication Interface	CAN、CANFD

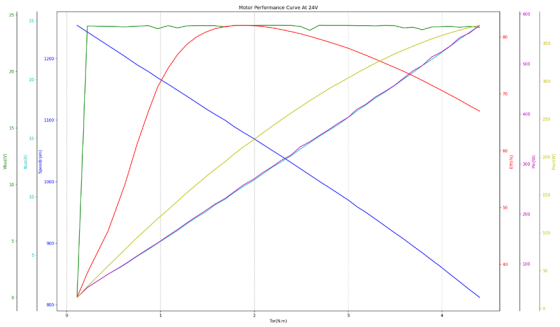
Dimension

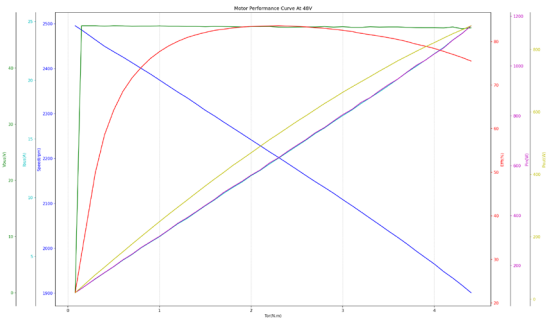






Performance Curve



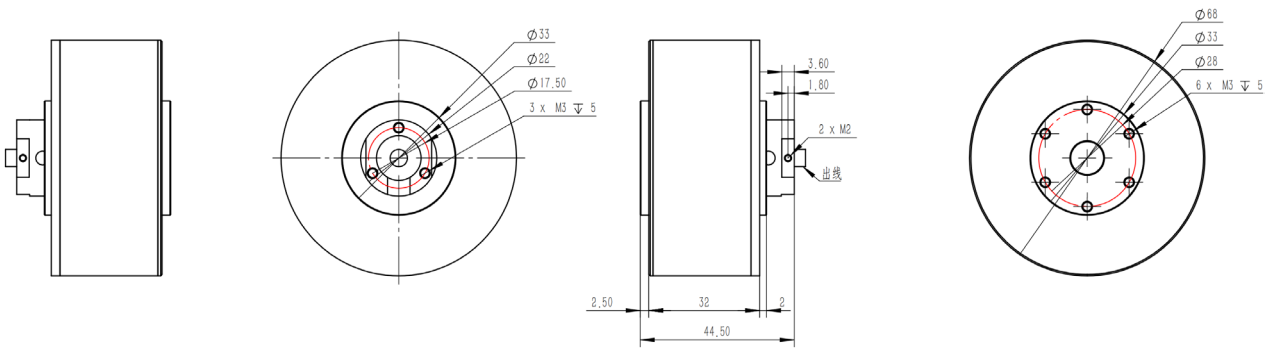


Hub Motor DM-H6215

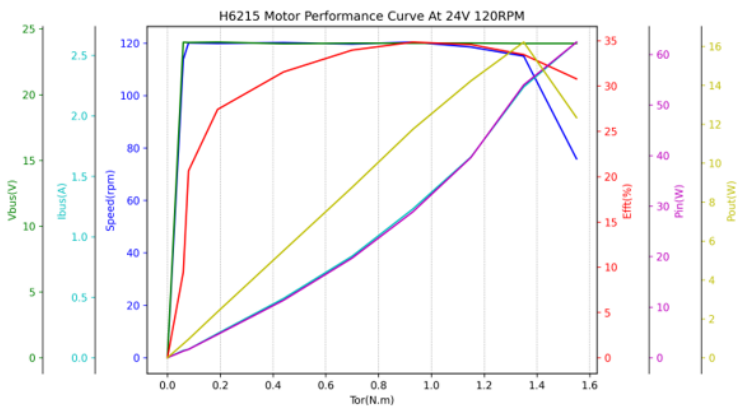


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	2.5
Peak Phase Current(A)	4.3
Rated Torque(Nm)	1
Peak Torque(Nm)	2
Rated Speed(rpm)	120
No-load Speed(rpm)	320
Rated Power(W)	12
Gear Ratio	1
Pair of Poles	14
Phase-to-Phase Inductance(uH)	2000
Phase-to-Phase Resistance(Ω)	2.5
Motor Size Diameter*Height(mm)	68*42.5
Weight (g)	360
Encoder	Hall Encoder Incremental Encoder
Communication Interface	CAN

Dimension



Performance Curve

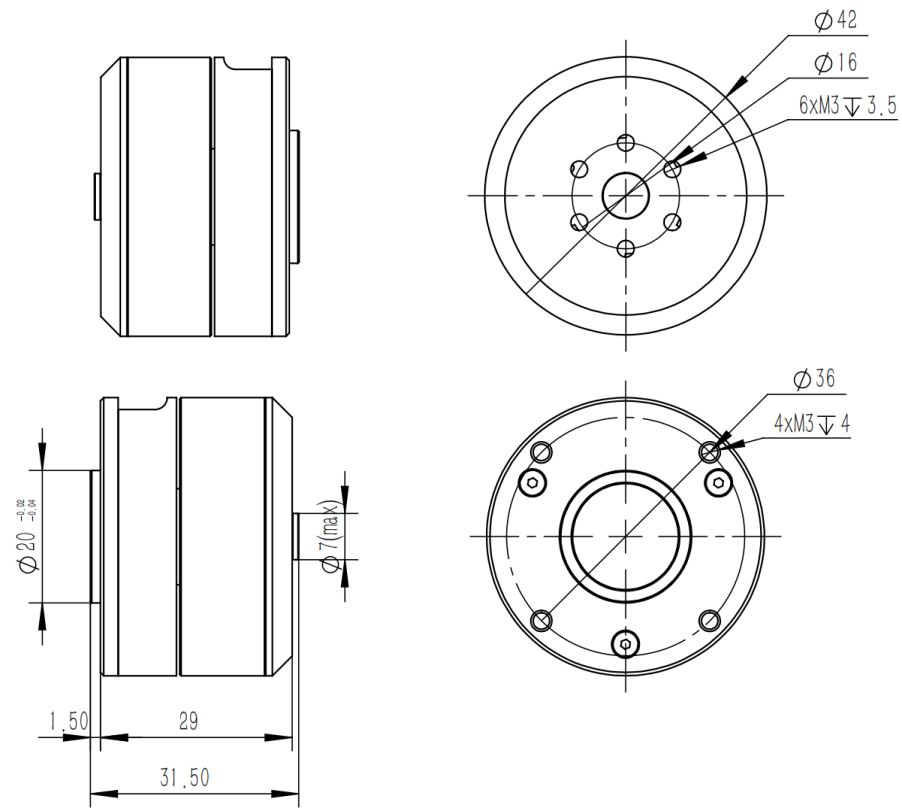


Hub Motor DM-H3510



Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	0.79
Peak Phase Current(A)	1.88
Rated Torque(Nm)	0.1
Peak Torque(Nm)	0.454
Rated Speed(rpm)	500
No-load Speed(rpm)	1700
Rated Power(W)	5
Gear Ratio	1
Pair of Poles	14
Phase-to-Phase Inductance(uH)	470
Phase-to-Phase Resistance(Ω)	2.76
Motor Size Diameter*Height(mm)	42*31.5
Weight (g)	116
Encoder	Magnetic Encoder Absolute
Communication Interface	CAN、CANFD

Dimension

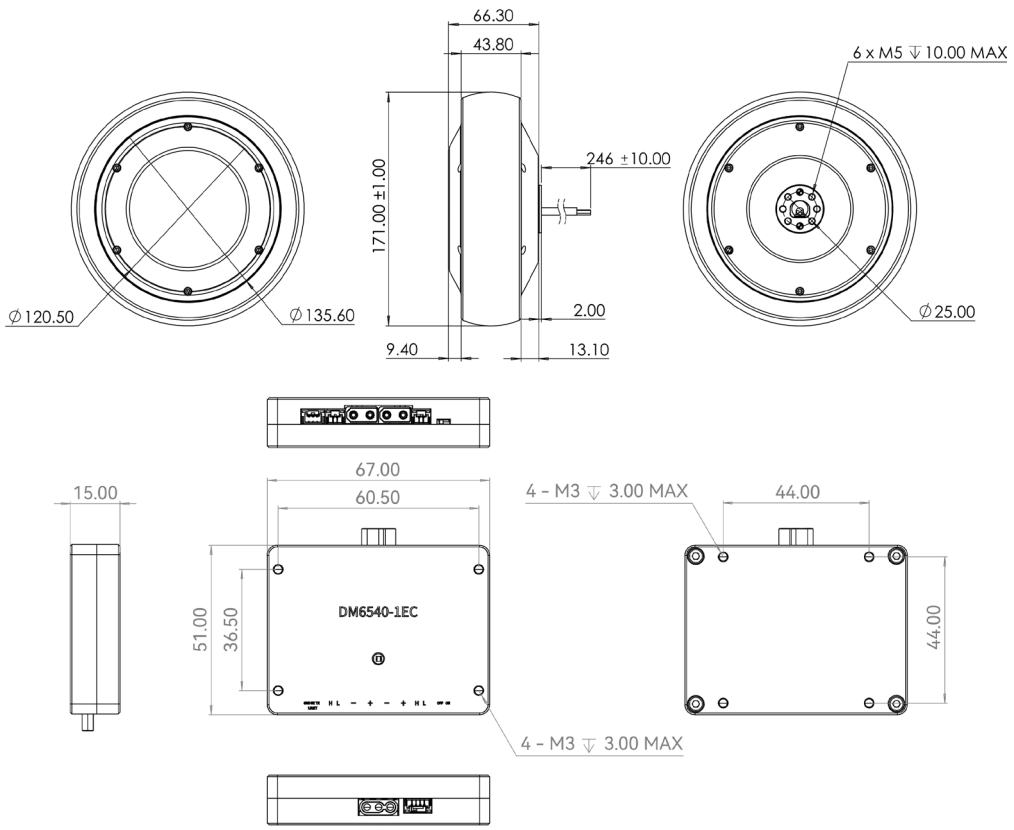


Hub Motor DM-H65

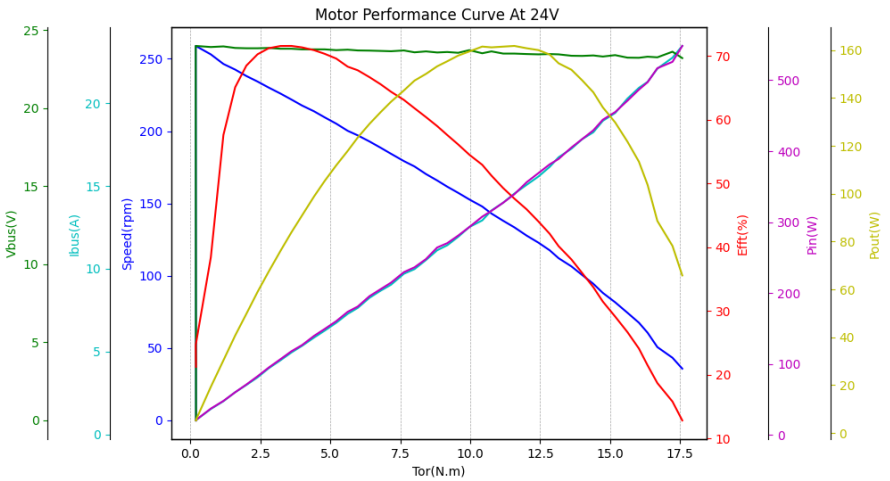


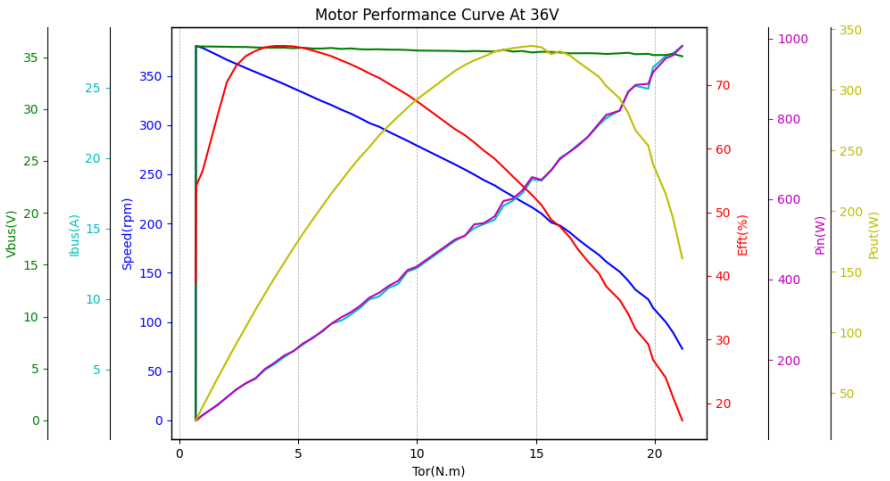
Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	9.4
Peak Phase Current(A)	33.6
Rated Torque(Nm)	6
Peak Torque(Nm)	21.5
Rated Speed(rpm)	120
No-load Speed(rpm)	260@24V ; 395@36V 480@48V
Rated Power(W)	250（6nm 400rpm @48V S2 Duty）
Gear Ratio	1
Pair of Poles	15
Phase-to-Phase Inductance(uH)	750
Phase-to-Phase Resistance(Ω)	0.42
Motor Size Diameter*Height(mm)	171*66.3
Weight (g)	3150
Encoder	Inductance Absolute
Communication Interface	CAN、CANFD

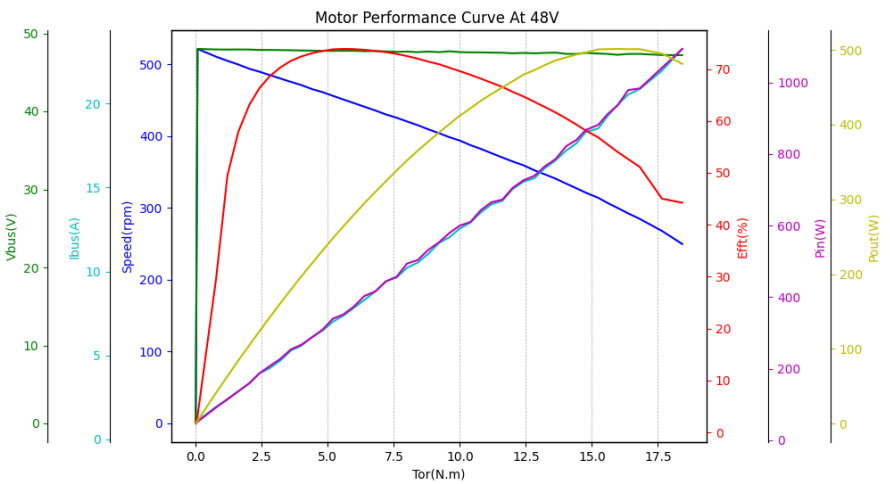
Dimension



Hub Motor DM-H65 Performance Curve





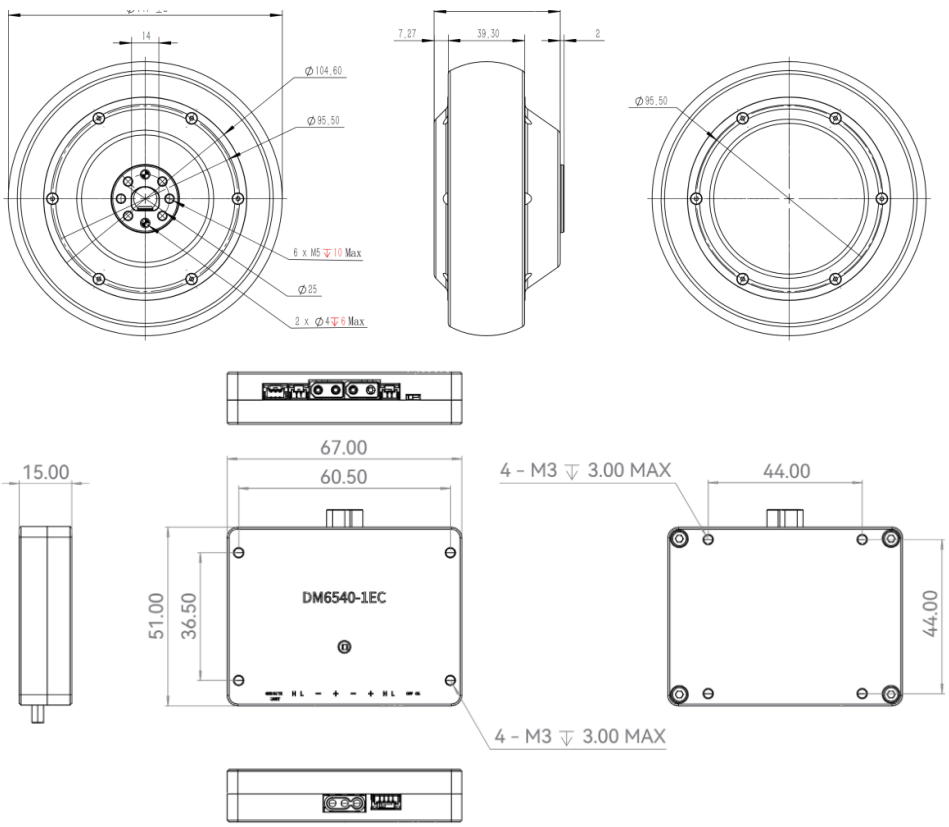


Hub Motor DM-H55

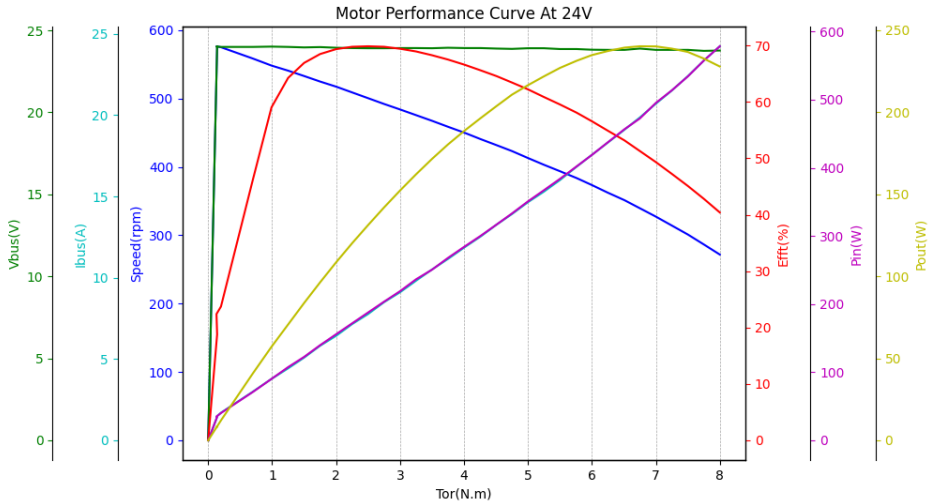


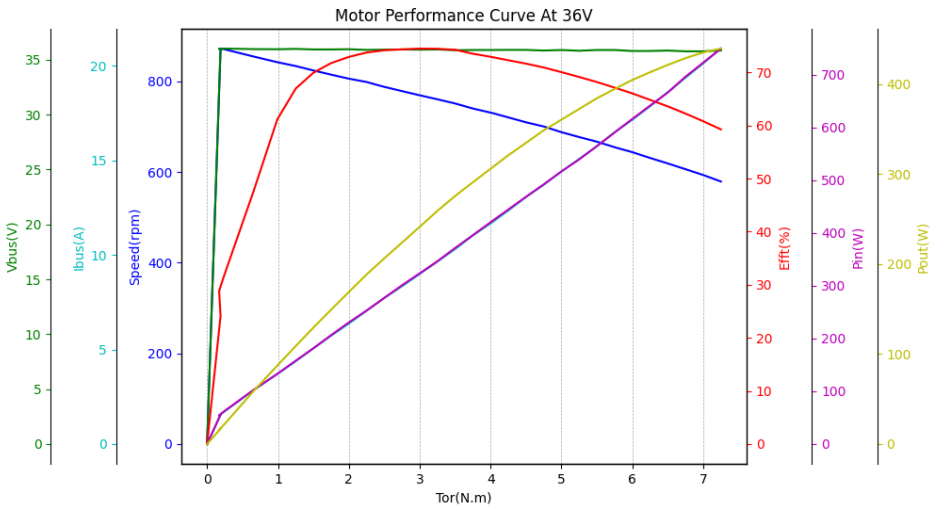
Specifications	
Supply Voltage(V)	24-48
Rated Phase Current(A)	13.05
Peak Phase Current(A)	33.55
Rated Torque(Nm)	3.5
Peak Torque(Nm)	9
Rated Speed(rpm)	350
No-load Speed(rpm)	587@24V ; 883@36V 1185@48V
Rated Power(W)	128
Gear Ratio	1
Pair of Poles	10
Phase-to-Phase Inductance(uH)	336
Phase-to-Phase Resistance(Ω)	0.281
Motor Size Diameter*Height(mm)	141*66.9
Weight (g)	2100
Encoder	Inductance Absolute
Communication Interface	CAN、CANFD

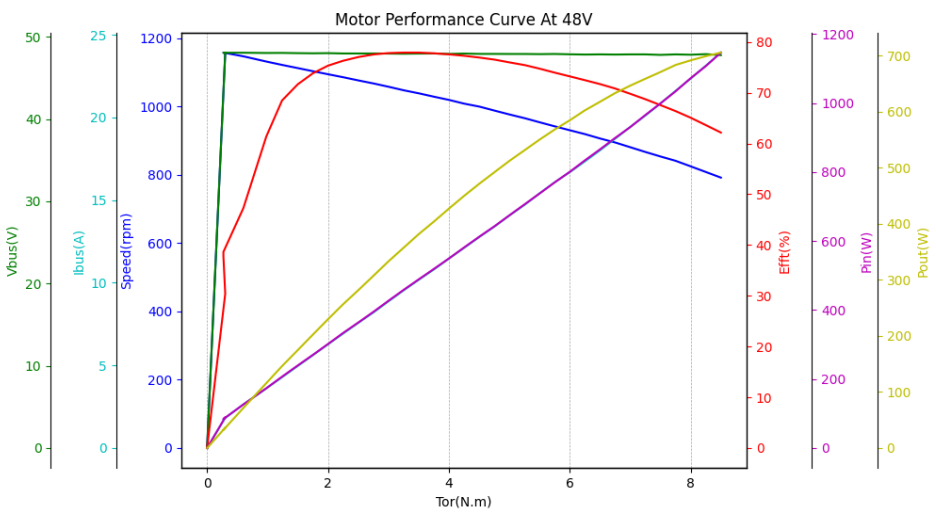
Dimension



Hub Motor DM-H55 Performance Curve





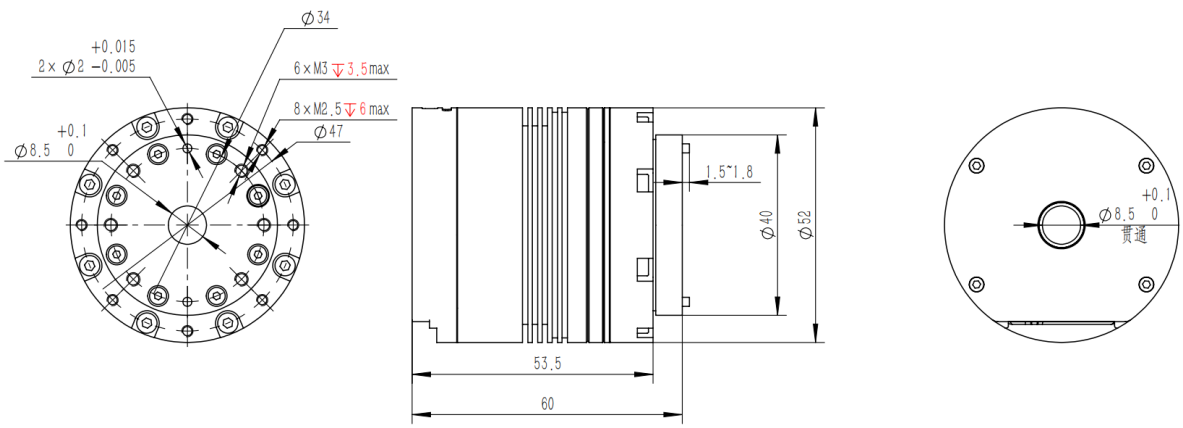


Harmonic Drive Joint Motor DM-JH11-51-2EC



Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	2.23
Peak Phase Current(A)	4.26
Rated Torque(Nm)	3.2
Peak Torque(Nm)	7.8
Rated Speed(rpm)	40
No-load Speed(rpm)	60
Rated Power(W)	13
Gear Ratio	51
Pair of Poles	7
Phase-to-Phase Inductance(uH)	538
Phase-to-Phase Resistance(Ω)	1.1
Motor Size Diameter*Height(mm)	52*60
Weight (g)	345
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension

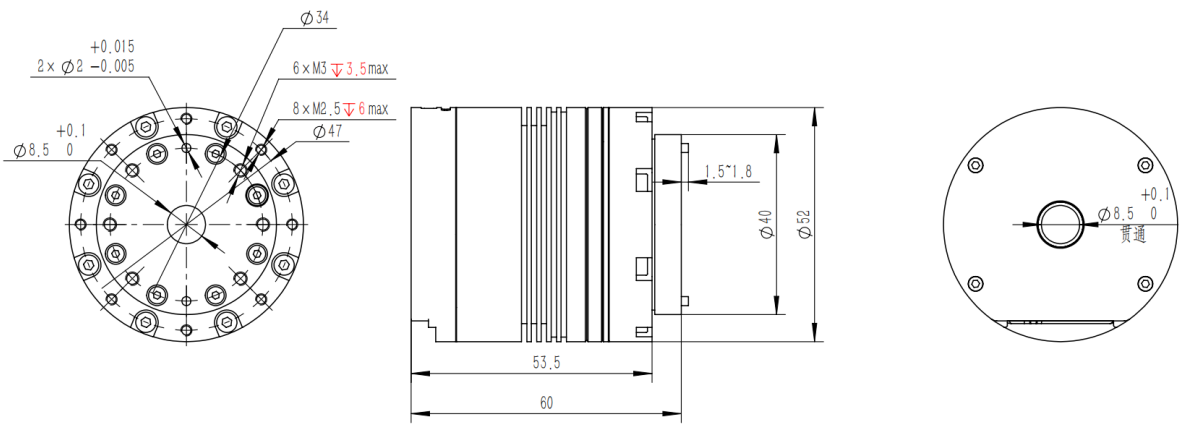


Harmonic Drive Joint Motor DM-JH11-101-2EC



Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	2.3
Peak Phase Current(A)	3.6
Rated Torque(Nm)	4.8
Peak Torque(Nm)	10.5
Rated Speed(rpm)	20
No-load Speed(rpm)	30
Rated Power(W)	10
Gear Ratio	101
Pair of Poles	7
Phase-to-Phase Inductance(uH)	538
Phase-to-Phase Resistance(Ω)	1.1
Motor Size Diameter*Height(mm)	52*60
Weight (g)	345
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension

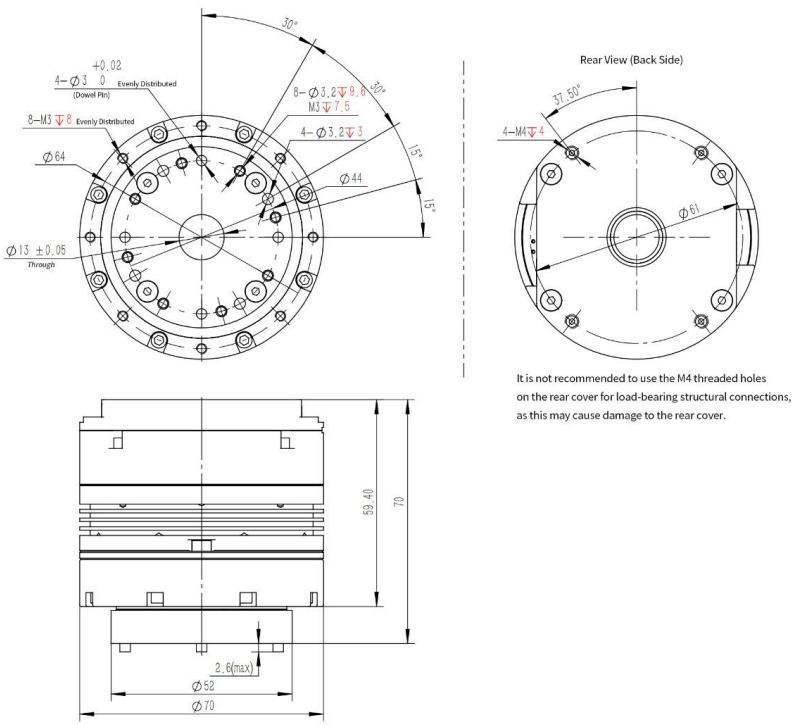


Harmonic Drive Joint Motor DM-JH14-51-2EC



Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	4.89
Peak Phase Current(A)	13.7
Rated Torque(Nm)	5.4
Peak Torque(Nm)	18
Rated Speed(rpm)	40
No-load Speed(rpm)	60
Rated Power(W)	22
Gear Ratio	51
Pair of Poles	10
Phase-to-Phase Inductance(uH)	170
Phase-to-Phase Resistance(Ω)	0.243
Motor Size Diameter*Height(mm)	70*70
Weight (g)	775
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension

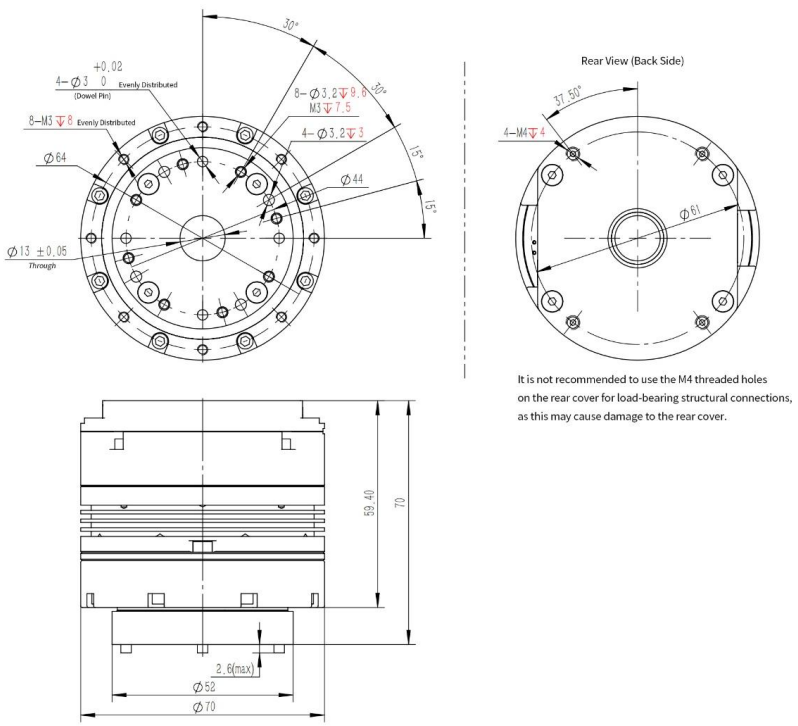


Harmonic Drive Joint Motor DM-JH14-101-2EC



Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	4.12
Peak Phase Current(A)	10.97
Rated Torque(Nm)	7.8
Peak Torque(Nm)	28
Rated Speed(rpm)	20
No-load Speed(rpm)	30
Rated Power(W)	16
Gear Ratio	101
Pair of Poles	10
Phase-to-Phase Inductance(uH)	170
Phase-to-Phase Resistance(Ω)	0.243
Motor Size Diameter*Height(mm)	70*70
Weight (g)	775
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension

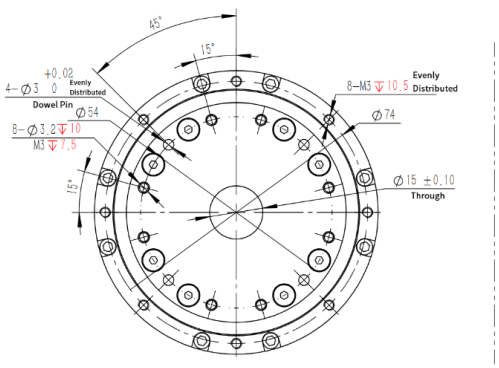


Harmonic Drive Joint Motor DM-JH17-51-2EC

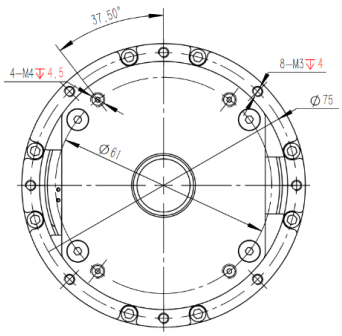


Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	4.62
Peak Phase Current(A)	12.94
Rated Torque(Nm)	16
Peak Torque(Nm)	24
Rated Speed(rpm)	40
No-load Speed(rpm)	60
Rated Power(W)	67
Gear Ratio	51
Pair of Poles	10
Phase-to-Phase Inductance(uH)	107
Phase-to-Phase Resistance(Ω)	0.161
Motor Size Diameter*Height(mm)	80*70
Weight (g)	985
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

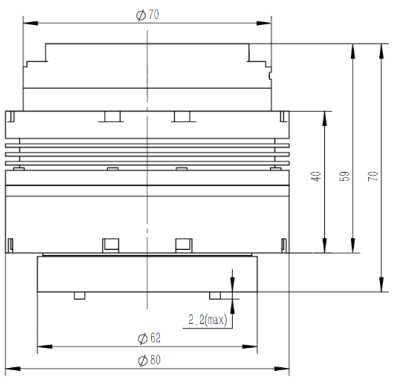
Dimension



Rear View (Back Side)



It is not recommended to use the M4 threaded holes on the rear cover for load-bearing structural connections, as this may cause damage to the rear cover.

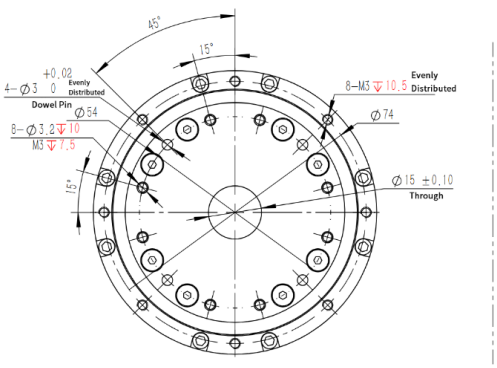


Harmonic Drive Joint Motor DM-JH17-101-2EC

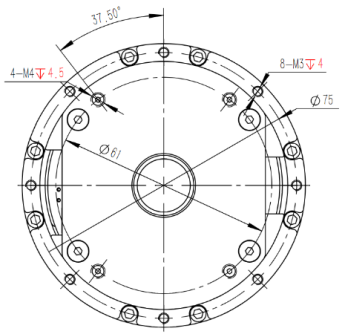


Specifications	
Supply Voltage(V)	24V (Support 24-48V)
Rated Phase Current(A)	6.31
Peak Phase Current(A)	16.85
Rated Torque(Nm)	24
Peak Torque(Nm)	54
Rated Speed(rpm)	20
No-load Speed(rpm)	30
Rated Power(W)	50
Gear Ratio	101
Pair of Poles	10
Phase-to-Phase Inductance(uH)	107
Phase-to-Phase Resistance(Ω)	0.161
Motor Size Diameter*Height(mm)	80*70
Weight (g)	985
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

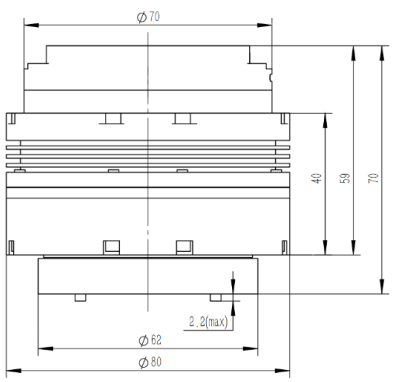
Dimension



Rear View (Back Side)



It is not recommended to use the M4 threaded holes on the rear cover for load-bearing structural connections, as this may cause damage to the rear cover.

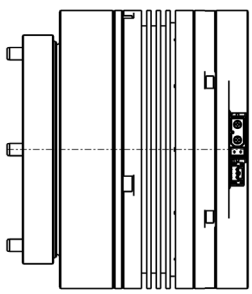


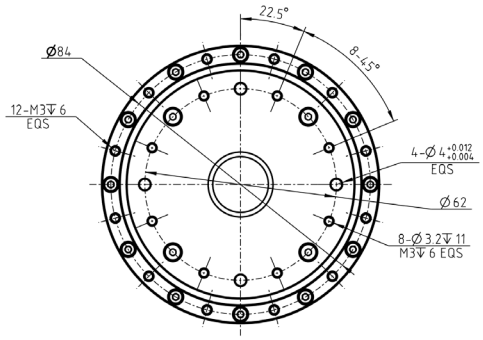
Harmonic Drive Joint Motor DM-JH20-51-2EC

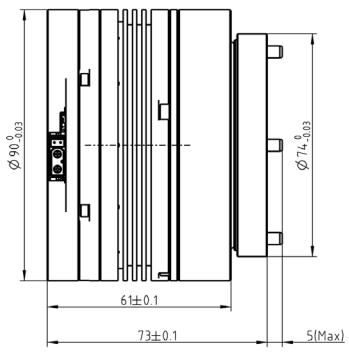


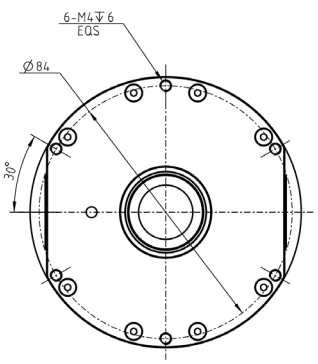
Specifications	
Supply Voltage(V)	48
Rated Phase Current(A)	6.42@24V 6.3@48V
Peak Phase Current(A)	13.2@24V 13.7@48V
Rated Torque(Nm)	32
Peak Torque(Nm)	69
Rated Speed(rpm)	40
No-load Speed(rpm)	60
Rated Power(W)	134
Gear Ratio	51
Pair of Poles	10
Phase-to-Phase Inductance(uH)	163.3
Phase-to-Phase Resistance(Ω)	0.2
Motor Size Diameter*Height(mm)	90*73
Weight (g)	1428
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension









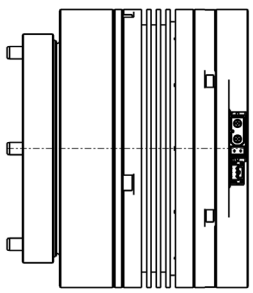
It is not recommended to use the M4 threaded hole on the back cover as a connection for load-bearing structural components, as it may cause damage to the back cover structure.

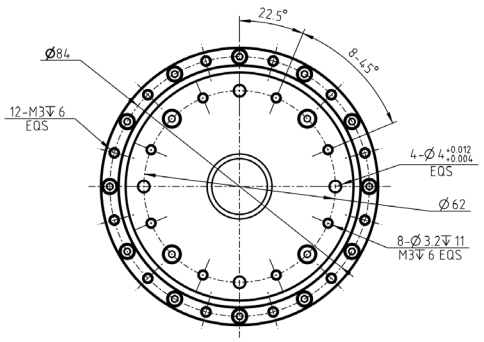
Harmonic Drive Joint Motor DM-JH20-101-2EC

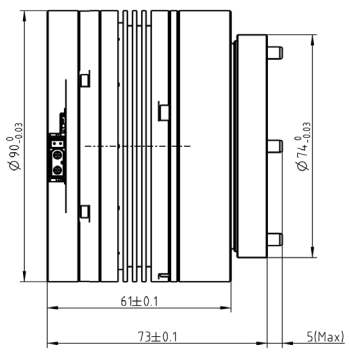


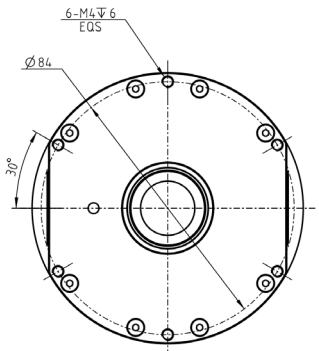
Specifications	
Supply Voltage(V)	48
Rated Phase Current(A)	3.52@24V 3.48@48V
Peak Phase Current(A)	10.3@24V 10.05@48V
Rated Torque(Nm)	32
Peak Torque(Nm)	102
Rated Speed(rpm)	20
No-load Speed(rpm)	30
Rated Power(W)	67
Gear Ratio	101
Pair of Poles	10
Phase-to-Phase Inductance(uH)	163.3
Phase-to-Phase Resistance(Ω)	0.2
Motor Size Diameter*Height(mm)	90*73
Weight (g)	1434
Encoder	Inductive/Dual Encoder
Communication Interface	CAN、CANFD

Dimension



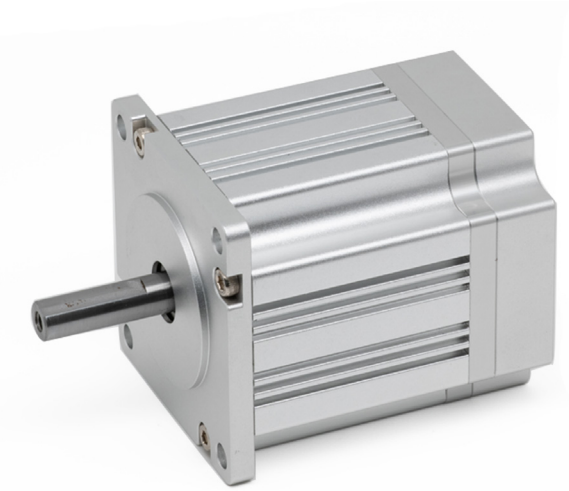






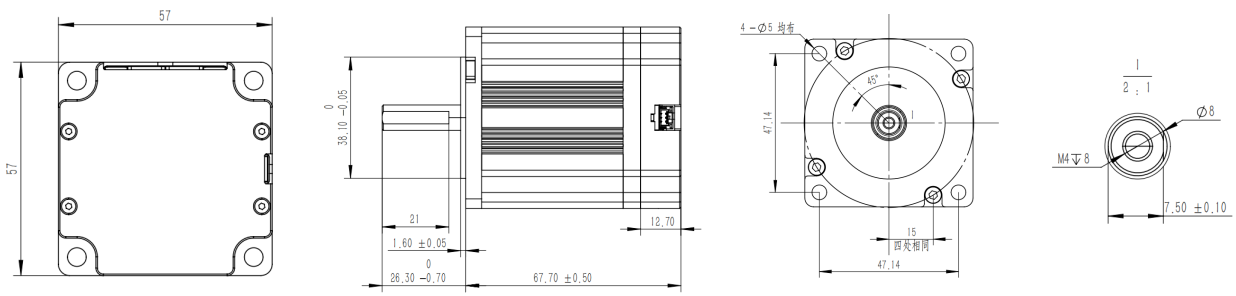
It is not recommended to use the M4 threaded hole on the back cover as a connection for load-bearing structural components, as it may cause damage to the back cover structure.

Direct Drive Motor DM-D5730-1EC

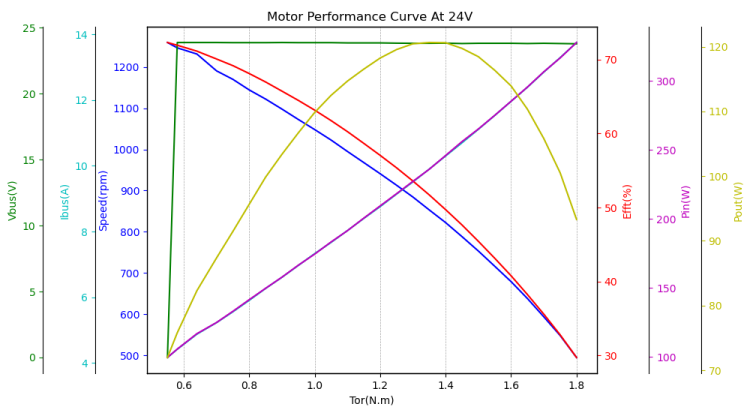


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	5.4
Peak Phase Current(A)	16.2
Rated Torque(Nm)	0.65
Peak Torque(Nm)	1.9
Rated Speed(rpm)	1100
No-load Speed(rpm)	1500
Rated Power(W)	74
Gear Ratio	1
Pair of Poles	14
Phase-to-Phase Inductance(uH)	85
Phase-to-Phase Resistance(Ω)	0.17
Motor Size Diameter*Height(mm)	57*95.2
Weight (g)	628
Encoder	Inductance Absolute
Communication Interface	CAN、CANFD

Dimension



Performance Curve

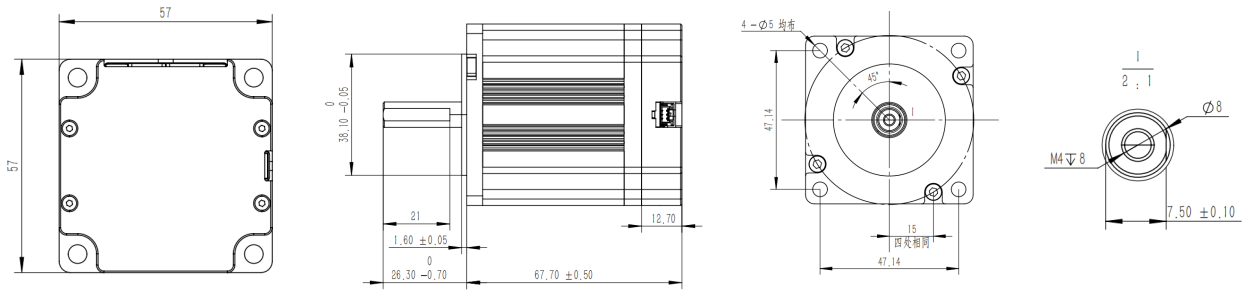


Direct Drive Motor DM-D5730-1EE

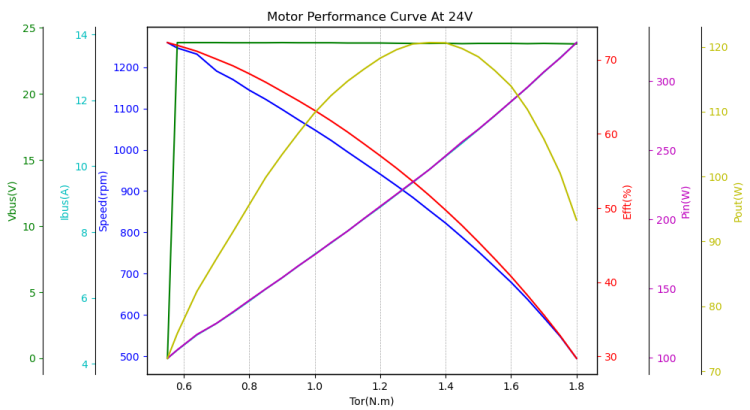


Specifications	
Supply Voltage(V)	24
Rated Phase Current(A)	5.4
Peak Phase Current(A)	16.2
Rated Torque(Nm)	0.65
Peak Torque(Nm)	1.9
Rated Speed(rpm)	1100
No-load Speed(rpm)	1500
Rated Power(W)	74
Gear Ratio	1
Pair of Poles	14
Phase-to-Phase Inductance(uH)	85
Phase-to-Phase Resistance(Ω)	0.17
Motor Size Diameter*Height(mm)	57*95.2
Weight (g)	628
Encoder	Inductance Absolute
Communication Interface	EtherCAT、CAN、CANFD

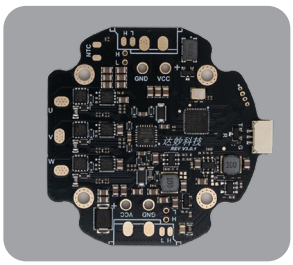
Dimension



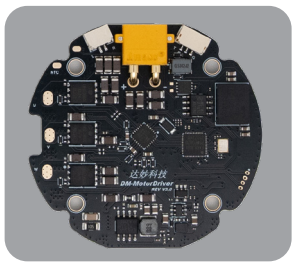
Performance Curve



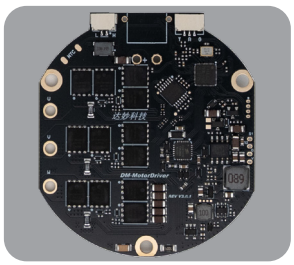
Driver



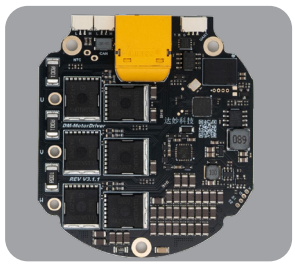
DM40-1E



DM60-1E



DM80-1E



DM100-1E

Name	DM40-1E	DM60-1E
Supply Voltage	24V (4S-6S) absolute max. 32 V	24V (4S-12S) absolute max. 85 V
MCU Core	Cortex®-M4	Cortex®-M4
MCU Clock Speed	200MHz	200MHz
Communications	CAN (UART for parameter configuration)	CAN (UART for parameter configuration)
Control Method	FOC	FOC
Maximum Continuous Current	4A	10A
Peak Current	10A	20A
Sensor Type	Magnetic encoder (dual magnetic encoder, single magnetic encoder + Hall)	Magnetic encoder (dual magnetic encoder, single magnetic encoder + Hall)
Control Modes	MIT、Speed、Position	MIT、Speed、Position
Encoder Interface	Supports dual encoders	Supports dual encoders
Compatible Motor Size: Stator Diameter(mm)	35-43	60

Name	DM80-1E	DM100-1E
Supply Voltage	24V (4S-12S) absolute max. 85 V	24V (4S-12S) absolute max. 85 V
MCU Core	Cortex®-M4	Cortex®-M4
MCU Clock Speed	200MHz	200MHz
Communications	CAN (UART for parameter configuration)	CAN (UART for parameter configuration)
Control Method	FOC	FOC
Maximum Continuous Current	20A	40A
Peak Current	40A	80A
Sensor Type	Magnetic encoder (dual magnetic encoder, single magnetic encoder + Hall)	Magnetic encoder (dual magnetic encoder, single magnetic encoder + Hall)
Control Modes	MIT、Speed、Position	MIT、Speed、Position
Encoder Interface	Supports dual encoders	Supports dual encoders
Compatible Motor Size: Stator Diameter(mm)	80	100

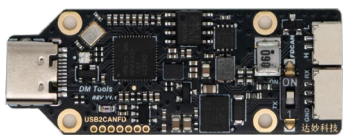
Development Kit Motor Setup Kit & Configuration Software

USB2CAN



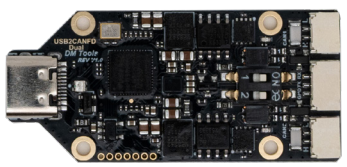
1、64 KB buffer, no data loss for large data streams;
2、Features CAN isolation;
3、CAN analyzer function with timed transmission;
4、Supports baud rates from 5 to 1000 KBPS;
5、Supports MIT motor debugging;
6、Supports USB-to-UART functionality;
7、Ultra-compact size, 39 * 18 mm.

USB2CANFD



1、128 KB buffer, no data loss for large data streams;
2、2Features CAN isolation;
3、CAN analyzer function with timed transmission;
4、Supports baud rates from 5 to 5000 kbps;
5、Supports MIT motor debugging;
6、Supports USB-to-UART conversion;
7、Ultra-compact size: 48 * 18 mm.

USB2CANFD Dual

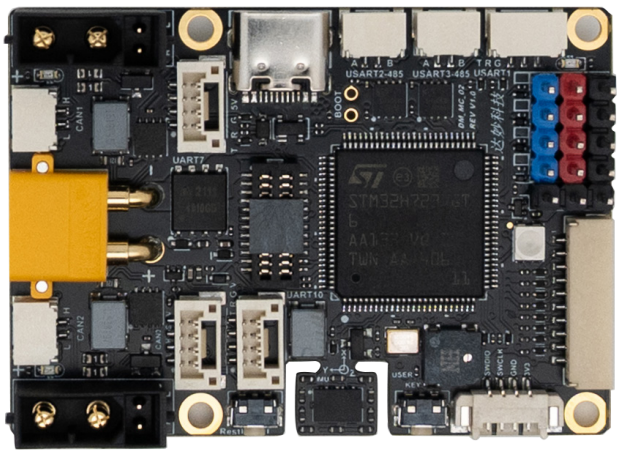


1、128KB ultra-large buffer, no data loss even at millions of frames;
2、Features CAN isolation;
3、Includes CAN bus analyzer functionality with support for scheduled transmission;
4、Supports baud rates from 5 to 5000 kbps;
5、Supports MIT motor debugging;
6、Supports USB-to-UART conversion;
7、Ultra-compact size: 48 * 22 mm;
8、Supports the Linux ecosystem.

Configuration Software



1、Supports visual calibration;
2、Enables real-time waveform display;
3、Facilitates easy parameter tuning;
4、Allows firmware updates;
5、Offers expanded test functions;
6、Adopts standardized operation logic;
7、Reduces the learning barrier for debugging;



H7 Control Board (DM-MC-Board02)

Based on STM32H723VGT6, designed for robotics control.

Features:

1.Reverse polarity protection;

2.Soft start;

3.Built-in high-performance IMU (BMI088).

Interfaces:

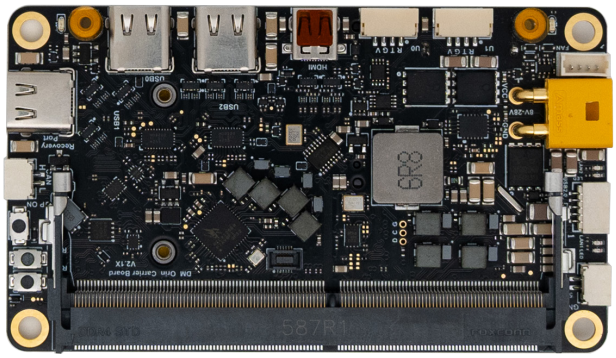
1. 3 × CAN FD;
2. 2 × RS485;
3. USB;
4. SBUS, PWM.

Applications:

1. AGV robots;
2. Self-balancing robots;
3. Legged robots;
4. DIY robotic systems.

Specifications

Mechanical	Supply Voltage	12~24V	Support 6S input
	Controllable Power Output	5 A per channel	—
	Onboard Power Output	5V@2A	External
	Weight	19.5g	—
	Dimensions	56x40mm	—
Technical	MCU	—	—
	Power switch	X2	—
	CANFD	X3	—
	RS485	X2	—
	USB	X1	—
	SBUS	X1	—
	PWM	X4	—
	Serial ports	X3	—
	SWD	X1	—
	W2812 RGB LED	X1	—
	Buzzer	X1	—
	Button	X2	User + Reset
	IMU	X1	BMI088
	LCD interface	X1	SPI+IIC+ADC
	QSPI Flash	8Mb	W25Q64JV
	DCMI Interface	X1	24-pin, no adapter required
	Expansion Interface	X1	2 UART 1SPI 1IIC



DM-ORIN NX V2.X

Compact carrier board for lightweight systems.

Compatible with:

1.NVIDIA® Jetson Orin NX;

2.Orin Nano.

Features:

1.Weight: 39.2 g;

2.Wide input voltage;

3.Supports SUPER mode;

4.Rich interfaces for compact applications.

Applications:

1.Drones;

2.Handheld mapping devices;

3.Legged robots.

Note: Xavier NX supports only one USB 3.0 port and no extended Ethernet.

Specifications			
Serial Number	Name	Interface Type	Note
1	Weight	39.2g	—
2	Supply Voltage	12~28V	Supports 6S input
3	Dimensions	50x86.8x13mm	—
4	Operating Temperature	-20°C to 50°C	—

Specifications				
Serial Number	Interface Name	Interface Type	Quantity	Note
1	USB3.0	TypeC	3	One port supports Recovery
2	USB2.0	SH1.0(5pin)	2	—
3	HDMI	MicroHDMI	1	—
4	Network Port	SH1.0(8pin)	1	On-board transformer
		SH1.0(8pin)	1	On-board transformer (supports PTP up to 2.5G)
5	M.2	M.2 KeyM	1	2230 SSD
		M.2 KeyE	1	2230 Network card (AW-CB375NF)
6	CAN	GH1.25 2pin	1	Supports up to 5Mbps without terminal resistor
7	Serial port	GH1.25 4pin	2	3.3V Level 5V Power Supply
8	CIS	0.5mmFPC 22Pin	2	Wire sequence is consistent with the original version, supporting top-and-bottom connection.
9	GPIO	0.5mmFPC 12Pin	1	SPI1、IIC1*、GPIO07 3.3V Level
10	Cooling Fan	1.25 4pin	1	Consistent with the original version
11		GH1.25 2pin	1	Provides only 5V power supply
12	Button	Turtle-shaped Tactile Switch	3	ReSet、Recovery、PowerON
13	RTC BAT	MY1220	1	CR1220 Button Cell Battery

Development Kit IMU Module

DM-IMU-L1



IMU with built-in BMI088 (3-axis accelerometer + 3-axis gyroscope).

Features:

1.Internal EKF for orientation output;

2.MCU up to 200 MHz;

3.Interfaces: USB, RS485, CAN;

4.Outputs:

- Acceleration
- Angular velocity
- Euler angles
- Quaternion
- Output rate: up to 1 kHz

Key Features:	Design:
1. CAN / USB / RS485 output;	1.Isolated sensor structure for vibration resistance;
2. Adjustable output rate (100–1000 Hz);	2.Built-in heating for temperature stability;
3. Temperature control via heating circuit;	3.Reduced gyro drift without full temperature calibration.

Attitude Performance			
Parameters		Typical value	(After calibration) Zero drift for one hour
Roll Angle	Range	±180°	0.02°
Yaw Angle	Range	±180°	12.686°
Pitch Angle	Range	±90°	0.064°

*Note: All measurements are taken at room temperature of 25° C.

Accelerometer	
Parameters	Typical value
Range	±6G
Degree of Freedom	3
Nonlinearity	±0.5 %FS
Zero Offset	20mg
Bandwidth	230Hz

Gyroscope	
Parameters	Typical value
Range	±2000 ° /s
Degree of Freedom	3
Nonlinearity	±0.05 %FS
Zero Offset	±1 ° /s

Physical and Electrical	
Parameters	Typical value
Input voltage	5-28V
Power Consumption	@24V: 2.5W during temperature rise, 0.47W when stable
Communication Interface	USB、RS485、CAN
Output Frequency	Adjustable from 100 to 1000 Hz
Dimensions	36x26x9mm

*Tips:

1. Acceleration and angular velocity specifications are based on the BMI088 data sheet. This IMU performs only data calibration without filtering, directly outputting calibrated values. For detailed parameters, refer to the MEMS chip manual.

2. Power consumption is ~0.34 W at 5 V input.

Shenzhen Damiao Technology Co., Ltd.

Address: Room409, University Town Entrepreneurship Park, No.10 Lishan Road, Pingshan Community, Taoyuan Subdistrict, Nanshan District Shenzhen, Guangdong, China

Phone:+8613560764159 (weChat: same number)

Email: zhouchengyan@mdmbot.com